WSQ COURSE SLIDES

3D Modelling with Blender for Beginners

Course Code: TGS-2022015368

Tertiary Infotech Academy Pte. Ltd.
UEN 201200696W

Trainer: _____

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COURSE ADMINISTRATION

00

Welcome, outcomes and assessment

Two days · 16 hours · Blender 5.2 LTS

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Digital Attendance (Mandatory)

1

AM

Complete the SSG digital attendance at the start of each day.

2

PM

Complete the afternoon attendance after lunch.

3

Assessment

Complete the assessment attendance before the WA and PP.

4

Funding

At least 75% attendance and a Competent result are required.

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About the Trainer



General Trainer

Complete this profile before class

NAME _____

TITLE / DESIGNATION _____

QUALIFICATIONS _____

AREAS OF EXPERTISE _____

TRAINING AND INDUSTRY EXPERIENCE _____

CONTACT _____

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About the Trainer



Dr Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE	Principal Trainer and courseware lead
CERTIFICATION	Adult education and applied technology training
DELIVERS	AI, automation, data and digital design programmes
FOUNDER	Tertiary Infotech / Tertiary Courses
APPROACH	Concept-led teaching with evidence-based practice
CONTACT	enquiry@tertiaryinfotech.com

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Let’s know each other

1

You

Name, role and organisation.

2

Experience

Your exposure to Blender or 3D design.

3

Use case

A model or workflow you want to improve.

4

Outcome

One capability you want to demonstrate by Day 2.

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Ground rules

1

Participate

Ask questions and explain your decisions.

2

Respect

Give constructive feedback on work, not people.

3

Protect

Use only authorised, non-confidential inputs.

4

Verify

Treat generated output as unverified until checked.

5

Version

Save incremental copies before risky operations.

6

Evidence

Capture the process, not only the final render.

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TWO-DAY JOURNEY

Course architecture

1

Topic 1

Analyse the brief, budget geometry and control Blender.

2

Topic 2

Model, sculpt, inspect and refine a prototype.

3

Topic 3

Rig, animate, surface and present the prototype.

4

Topic 4

Light, render, optimise and export the asset.

5

Labs

Eight connected, evidence-led Blender labs.

6

Assessment

Nine SAQs and four integrated practical tasks.

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SCHEDULE

Lesson plan — two-day journey

DAY 1

- Creative briefs and polygon budgets
- Blender interface and scene control
- Modelling, sculpting and topology
- Labs 1–4

DAY 2

- Materials, rigging and animation
- Cameras, lighting and rendering
- Optimisation and export
- Labs 5–8 · WA + PP

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BY THE END OF THE COURSE

Learning outcomes

1

LO1

Analyse a creative brief and establish a suitable polygon budget.

2

LO2

Develop 3D models and prototypes in Blender.

3

LO3

Animate 3D models using suitable rigging, keyframe and presentation methods.

4

LO4

Optimise, light, render and export 3D models for the intended use.

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TSC alignment

1

K1–K2

Creative brief, budgets, interface and scene control.

2

K3–K4

Modelling, sculpting, topology and modifiers.

3

K5–K9

Animation, materials, review, rendering and export.

4

A1–A8

Brief, build, animate, present, optimise and deliver.

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READ BEFORE ASSESSMENT

Briefing for assessment

1

Individual

No discussion; complete your own submission.

2

Open book

Use slides, Learner Guide and approved materials only.

3

Devices

Place phones and other unapproved materials under the table.

4

Integrity

No photography or recording of assessment scripts.

5

Writing

Use black/blue pen where hard copy is used; no correction fluid.

6

Submission

Upload the completed document to the LMS.

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Final assessment

Written Assessment

- 9 open-ended SAQs
- K1–K9 coverage
- 60 minutes · open book

Practical Performance

- 4 integrated Blender tasks
- A1–A8 coverage
- 90 minutes · open book

Competency

- Assessment attendance
- Competent / Not Yet Competent
- Appeal and reassessment available

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ON ASSESSMENT DAY

Assessment flow

1. TRAQOM / SSG attendance
2. Written Assessment (SAQ)
3. Practical Performance (PP)
4. Upload answers to LMS
5. Sign Assessment Summary Record

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Access the lab packs

Option A

- Open the GitHub repository
- Clone with Git
- Work from labs/NN-title

Option B

- GitHub Code menu
- Download ZIP
- Extract before opening files

Evidence

- Follow INSTRUCTIONS.md
- Use CHECKLIST.md
- Update Example-Presentation.pptx

Open the GitHub lab repository

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From blockout to production-ready asset



Compare the same design at three maturity levels: blockout, topology-controlled model and finished material render. Each level answers a different review question.

Concept visual — validate against the editable Blender file.

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PROGRESS IS EARNED THROUGH EVIDENCE, NOT UNSUPPORTED PERCENTAGES

Production-readiness gates

1. Recognisable blockout
2. Controlled topology
3. Validated materials and motion
4. Target render and export
5. Clean round-trip handoff

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Source-informed build

1

Legacy v10

Preserved Blender foundations, modelling, sculpting, materials, animation and rendering.

2

Course page

Aligned the verified two-day structure and four learning outcomes.

3

Reference labs

Reused the supplied editable Blender sample files as hands-on inputs.

4

Blender Manual

Updated terminology and workflow for Blender 5.2 LTS.

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TOPIC 1

01

Get Started with Blender and the Creative Brief

Analyse requirements · establish budgets · control the scene

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PROGRESSIVE LEARNING

Topic 1 — capability map

1

Understand

Analyse requirements

2

Apply

establish budgets

3

Review

LO1 · K1–K2 · A1–A2

4

Demonstrate

Labs 1–2

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Read the creative brief as a production contract

1

Purpose

Audience, platform and viewing distance

2

Form

Silhouette, proportion and reference views

3

Constraints

Time, dimensions, format and budget

4

Evidence

Named checks, views and approval owner

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Decisions and risks — Read the creative brief as a production contract

DECISIONS THAT IMPROVE QUALITY

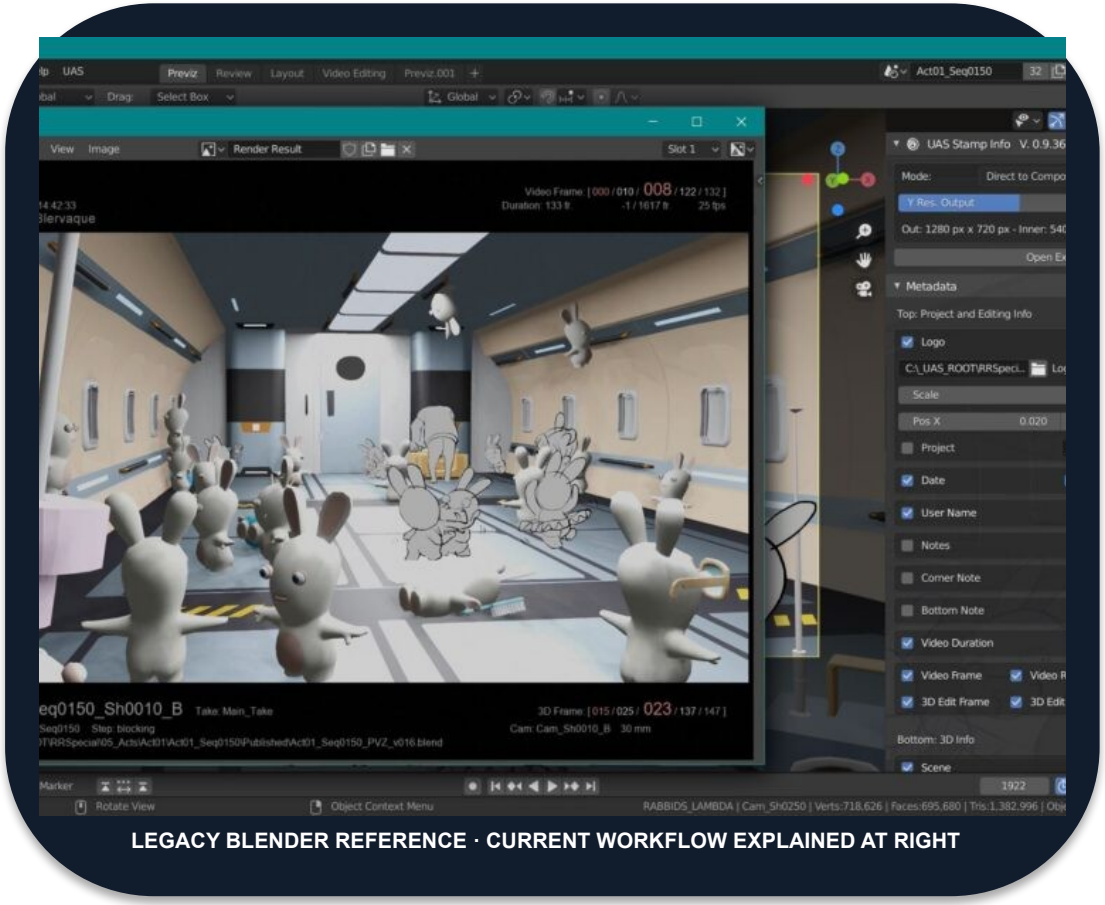
- Separate must-haves from preferences
- Resolve missing dimensions before modelling
- Name the final approver
- Freeze a versioned brief

FAILURE MODES TO CONTROL

- Starting from an attractive image instead of requirements
- Conflicting units or target platforms
- No measurable acceptance rule

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Read the creative brief as a production contract



PRINCIPLE

Turn subjective intent into testable form, function and delivery criteria.

APPLIED EXAMPLE

A goldfish hero asset needs a readable silhouette, animated fins, a 500-polygon target and a GLB handoff.

EVIDENCE TO TRACK

1. All mandatory fields resolved
2. Polygon target recorded
3. Approver and evidence named

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Set a polygon budget that matches the use

1. **Target**

   Mobile, web, film, still or print

2. **Density**

   Allocate loops to visible and moving regions

3. **LOD**

   Plan variants for different viewing conditions

4. **Measure**

   Use statistics and triangulated counts

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Decisions and risks — Set a polygon budget that matches the use

DECISIONS THAT IMPROVE QUALITY

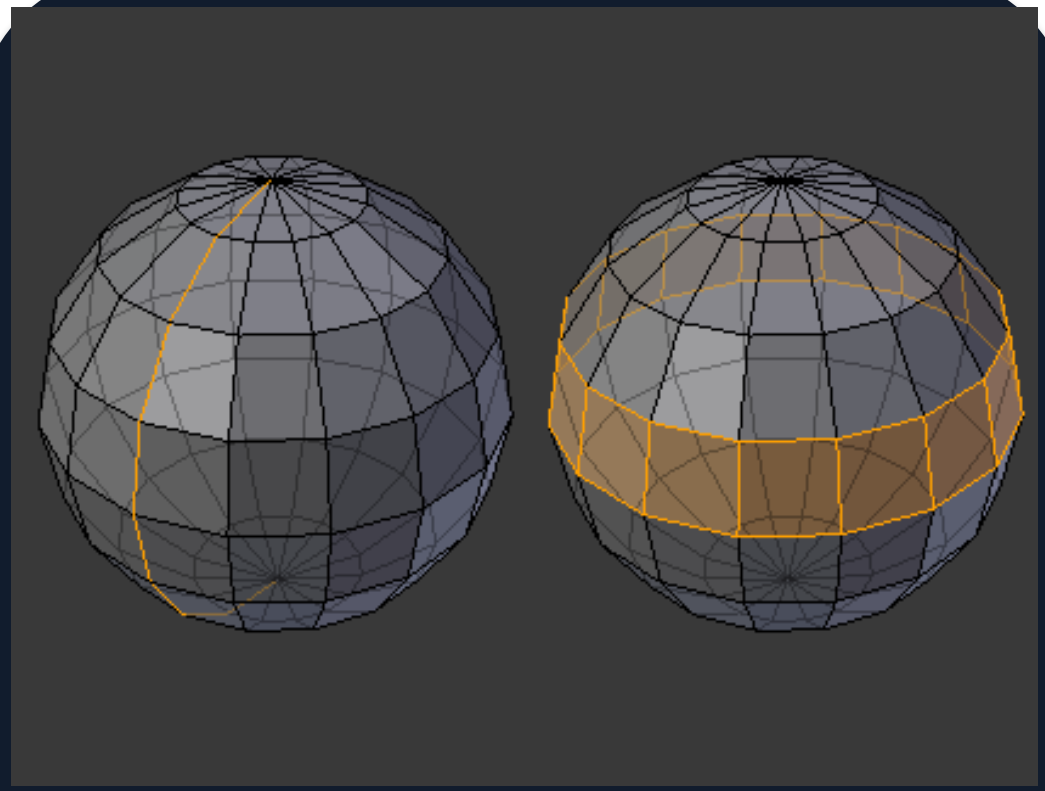
- Define whether the limit is faces or triangles
- Reserve loops around eyes, mouth and fin roots
- Keep a budget margin for revisions
- Check after modifiers

FAILURE MODES TO CONTROL

- Counting before unapplied modifiers
- Uniform density wastes geometry
- Budget reduction destroys silhouette

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Set a polygon budget that matches the use



LEGACY BLENDER REFERENCE · CURRENT WORKFLOW EXPLAINED AT RIGHT

PRINCIPLE

Spend geometry where it changes silhouette, deformation or shading.

APPLIED EXAMPLE

A 500-polygon fish keeps loops around the eye and tail while the body uses broad, clean quads.

EVIDENCE TO TRACK

1. Triangle count within target
2. Silhouette passes key views
3. Deformation zones retain support

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Understand Blender workspaces and editors

1

3D Viewport

Select, transform and inspect objects

2

Outliner

Control hierarchy, collections and visibility

3

Properties

Edit object, data, modifier and render settings

4

Timeline

Navigate frames and keyframes

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Decisions and risks — Understand Blender workspaces and editors

DECISIONS THAT IMPROVE QUALITY

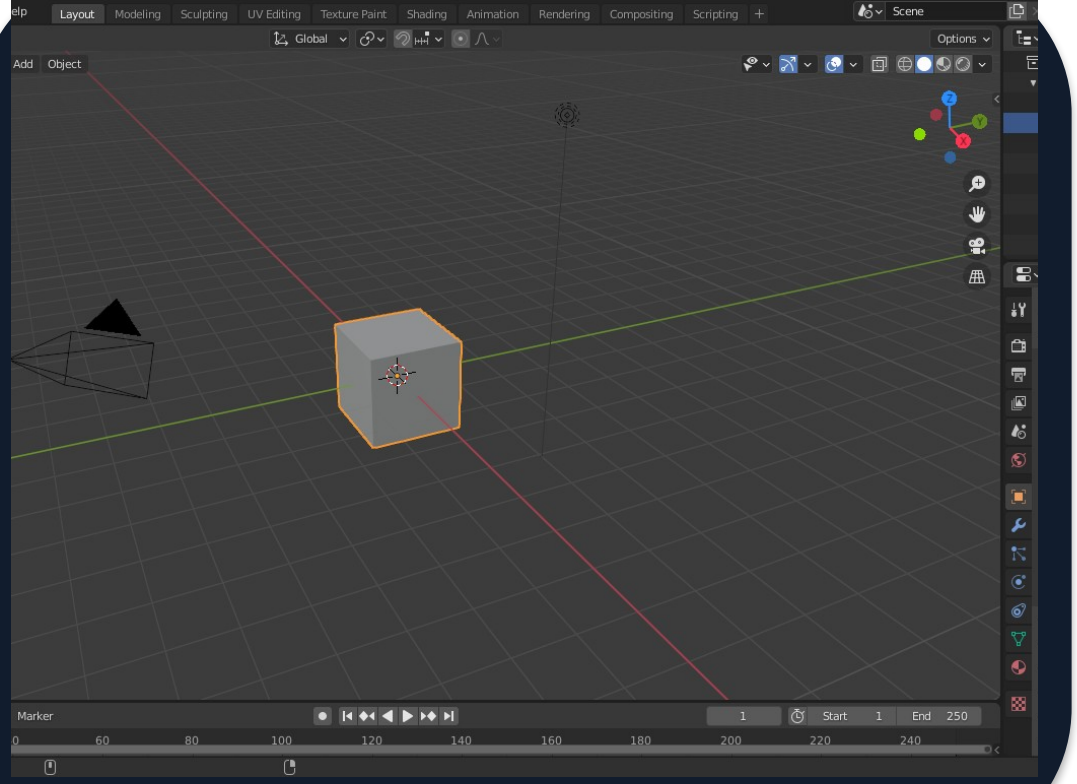
- Choose a workspace for the current task
- Keep the Outliner visible on complex scenes
- Confirm active object and mode
- Save layout changes only when useful

FAILURE MODES TO CONTROL

- Editing the wrong active object
- Hidden collection mistaken for missing work
- Wrong editor or mode causes commands to fail

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Understand Blender workspaces and editors



LEGACY BLENDER REFERENCE · CURRENT WORKFLOW EXPLAINED AT RIGHT

PRINCIPLE

Workspaces arrange editors; editors expose different views of the same data.

APPLIED EXAMPLE

The Layout workspace supports blockout; Sculpting and Shading workspaces expose specialised controls without changing the model data.

EVIDENCE TO TRACK

1. Correct object active
2. Correct mode visible
3. Scene hierarchy understandable

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Navigate, select and transform with intent

1

View

Orbit, pan, zoom and frame selection

2

Select

Object, component and hierarchy selection

3

Transform

Move, rotate and scale on constrained axes

4

Pivot

Median, cursor and individual origins

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Decisions and risks — Navigate, select and transform with intent

DECISIONS THAT IMPROVE QUALITY

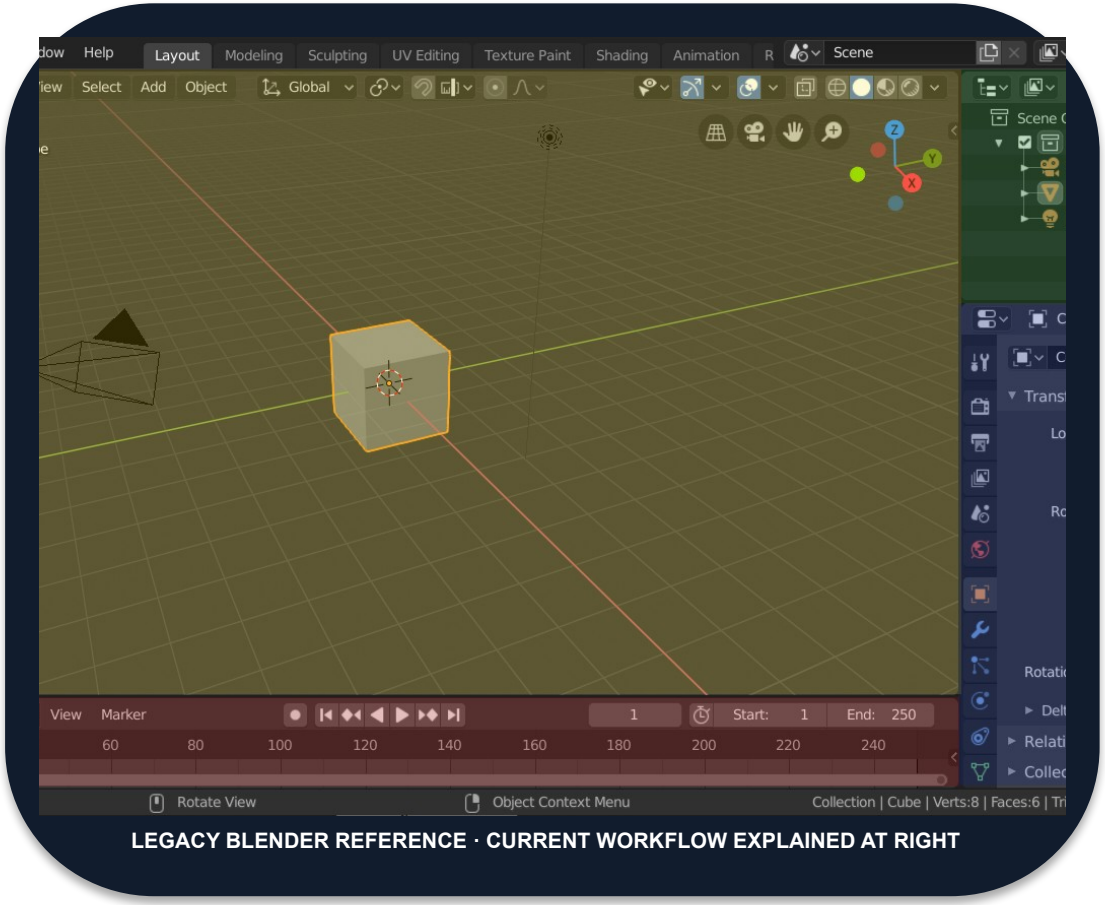
- Use orthographic views for proportion checks
- Constrain transformations to an axis
- Apply scale before scale-sensitive tools
- Place origins deliberately

FAILURE MODES TO CONTROL

- Unapplied non-uniform scale
- Accidental proportional editing
- Origin placement breaks rotation

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Navigate, select and transform with intent



LEGACY BLENDER REFERENCE · CURRENT WORKFLOW EXPLAINED AT RIGHT

PRINCIPLE

Reliable modelling begins with controllable views, pivots and transforms.

APPLIED EXAMPLE

Block a fish from side and top orthographic views, then inspect perspective for volume and proportion.

EVIDENCE TO TRACK

1. Transforms documented
2. Reference views aligned
3. No accidental offsets

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Organise objects, data and modifiers

1

Objects

Transform containers that reference data

2

Collections

Logical groups for visibility and export

3

Names

Role-based, consistent object/material labels

4

Modifiers

Ordered, non-destructive operations

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Decisions and risks — Organise objects, data and modifiers

DECISIONS THAT IMPROVE QUALITY

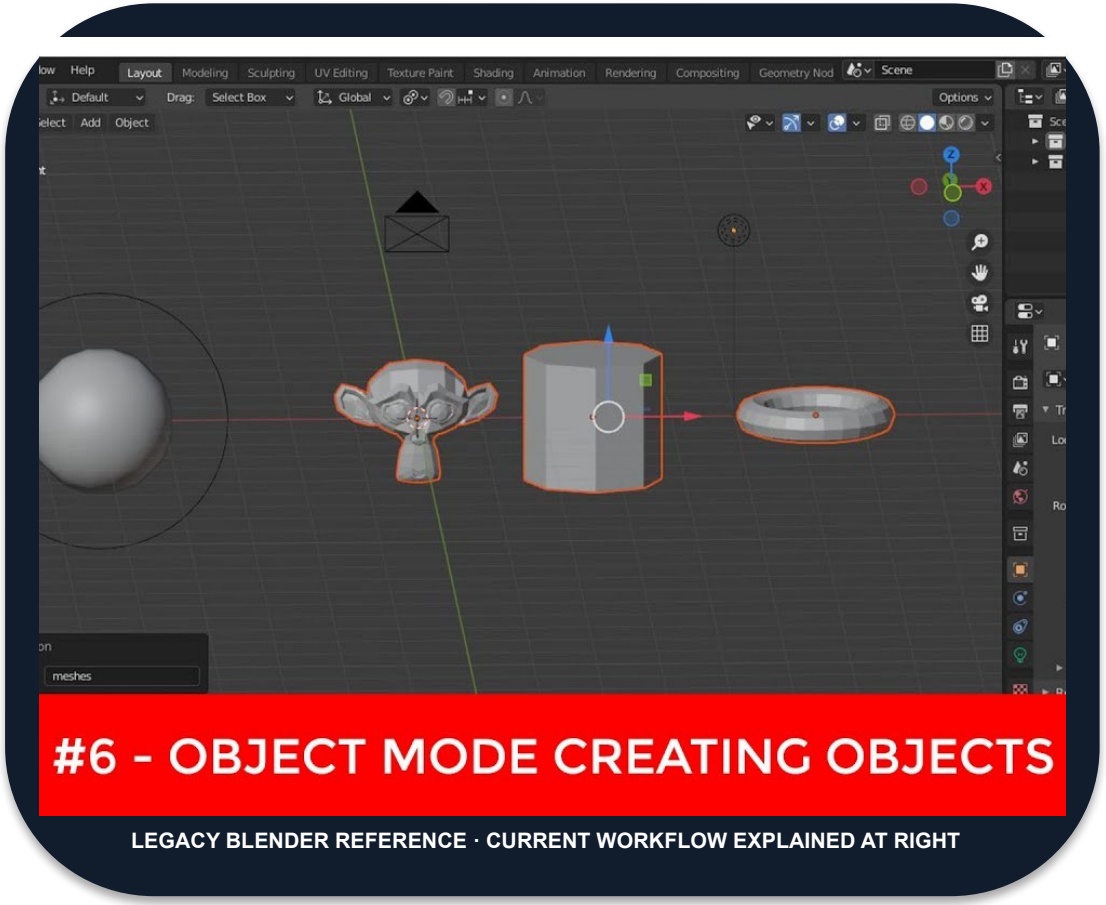
- Name objects before duplication
- Group exportable assets in one collection
- Keep a baseline before applying modifiers
- Order Mirror before Subdivision where appropriate

FAILURE MODES TO CONTROL

- Unclear Cube.001 naming
- Modifier order changes the result
- Applying too early removes editability

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Organise objects, data and modifiers



PRINCIPLE

A clean scene makes decisions reversible and handoffs understandable.

APPLIED EXAMPLE

The fish body, eyes and fins are named, collected and mirrored non-destructively before detail is added.

EVIDENCE TO TRACK

1. Zero ambiguous names
2. Hierarchy matches deliverable
3. Modifier intent recorded

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HANDS-ON PRACTICE

Topic 1 labs

Lab 1

- Navigate Blender and organise a production scene
- A named, organised scene with a short evidence presentation.

Lab 2

- Interpret a fish brief, budget polygons and build a blockout
- A brief, scorecard and editable goldfish blockout within the stated budget.

Progression

- Inspect and plan
- Build and refine
- Verify and present

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Navigate Blender and organise a production scene

LAB 1

You are onboarding to a small studio and must turn an unfamiliar Blender file into a controlled working scene.

DELIVERABLE

A named, organised scene with a short evidence presentation.

ALIGNMENT

K2 · A1

OBJECTIVE

Navigate the interface, distinguish Object and Edit modes, organise objects and preserve a clean baseline.

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Lab 1 — example evidence



Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.

Concept visual — validate against the editable Blender file.

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ACCEPTANCE CRITERIA

Lab 1 — quality gates

Form

- Required editors and modes are correctly identified
- Objects and collections use meaningful names

Structure

- Transforms are intentional and recorded
- Original starter remains unchanged

Delivery

- Saved file reopens without missing data
- A named, organised scene with a short evidence presentation.

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VERIFY, DOCUMENT, SUBMIT

Lab 1 — acceptance evidence

ACCEPTANCE CHECKS

- Required editors and modes are correctly identified
- Objects and collections use meaningful names
- Transforms are intentional and recorded
- Original starter remains unchanged
- Saved file reopens without missing data

EVIDENCE TO SUBMIT

- Interface screenshot
- Perspective and orthographic comparison
- Outliner hierarchy
- Transform panel
- Saved-version confirmation

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Interpret a fish brief, budget polygons and build a blockout

LAB 2

A client needs a recognisable goldfish prototype for a lightweight interactive scene.

DELIVERABLE

A brief, scorecard and editable goldfish blockout within the stated budget.

ALIGNMENT

K1–K2 · A1–A2

OBJECTIVE

Translate the brief into criteria, allocate a polygon budget and create a modifier-led blockout.

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Lab 2 — example evidence



Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.

Concept visual — validate against the editable Blender file.

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ACCEPTANCE CRITERIA

Lab 2 — quality gates

Form

- Brief contains measurable requirements
- Budget definition and allocation are explicit

Structure

- Blockout silhouette matches the concept
- Modifier workflow remains editable

Delivery

- Recorded count is within target or has a justified action
- A brief, scorecard and editable goldfish blockout within the stated budget.

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Lab 2 — acceptance evidence

ACCEPTANCE CHECKS

- Brief contains measurable requirements
- Budget definition and allocation are explicit
- Blockout silhouette matches the concept
- Modifier workflow remains editable
- Recorded count is within target or has a justified action

EVIDENCE TO SUBMIT

- Creative brief
- Polygon allocation table
- Three key views
- Modifier stack
- Viewport statistics

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WHAT YOU CAN NOW DEMONSTRATE

Topic 1 recap

1

Concepts

5 decision-ready frameworks

2

Labs

Labs 1–2 completed

3

Evidence

Editable Blender files, metrics and presentations

4

Alignment

LO1 · K1–K2 · A1–A2

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TOPIC 2

02

Develop 3D Models and Prototypes

Block out form · sculpt detail · refine topology

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PROGRESSIVE LEARNING

Topic 2 — capability map

1

Understand

Block out form

2

Apply

sculpt detail

3

Review

LO2 · K3–K4, K7 · A3–A4

4

Demonstrate

Labs 3–4

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Translate concept art into three-dimensional form

1

Views

Front, side, top and perspective references

2

Landmarks

Eye line, widest body, fin roots and tail

3

Volume

Primary, secondary and tertiary forms

4

Review

Compare silhouette before detail

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Decisions and risks — Translate concept art into three-dimensional form

DECISIONS THAT IMPROVE QUALITY

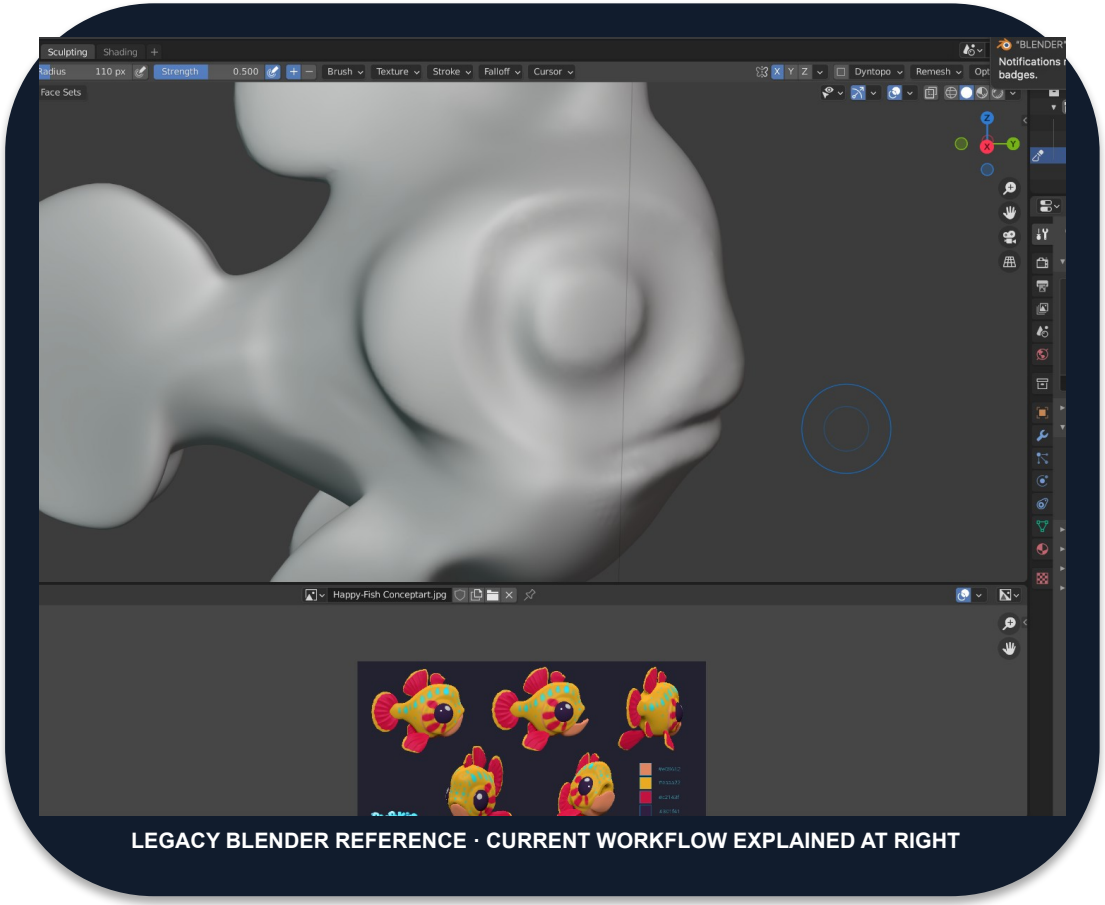
- Choose a dominant reference view
- Mark proportion landmarks
- Resolve contradictions explicitly
- Review at intended camera distance

FAILURE MODES TO CONTROL

- Perspective distortion in concept art
- Detail hides proportion errors
- Reference planes misaligned

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Translate concept art into three-dimensional form



LEGACY BLENDER REFERENCE · CURRENT WORKFLOW EXPLAINED AT RIGHT

PRINCIPLE

Use multiple views and landmark proportions; do not copy a single perspective literally.

APPLIED EXAMPLE

A fish blockout begins with body mass, tail plane and eye landmarks before scales or surface detail.

EVIDENCE TO TRACK

1. Landmarks align in key views
2. Silhouette recognisable
3. Primary volumes complete

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Model with mesh anatomy and edge flow

1

Components

Vertices, edges, faces and normals

2

Loops

Continuous paths around forms and joints

3

Topology

Quad-led structure with intentional poles

4

Normals

Face direction and smooth shading behaviour

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Decisions and risks — Model with mesh anatomy and edge flow

DECISIONS THAT IMPROVE QUALITY

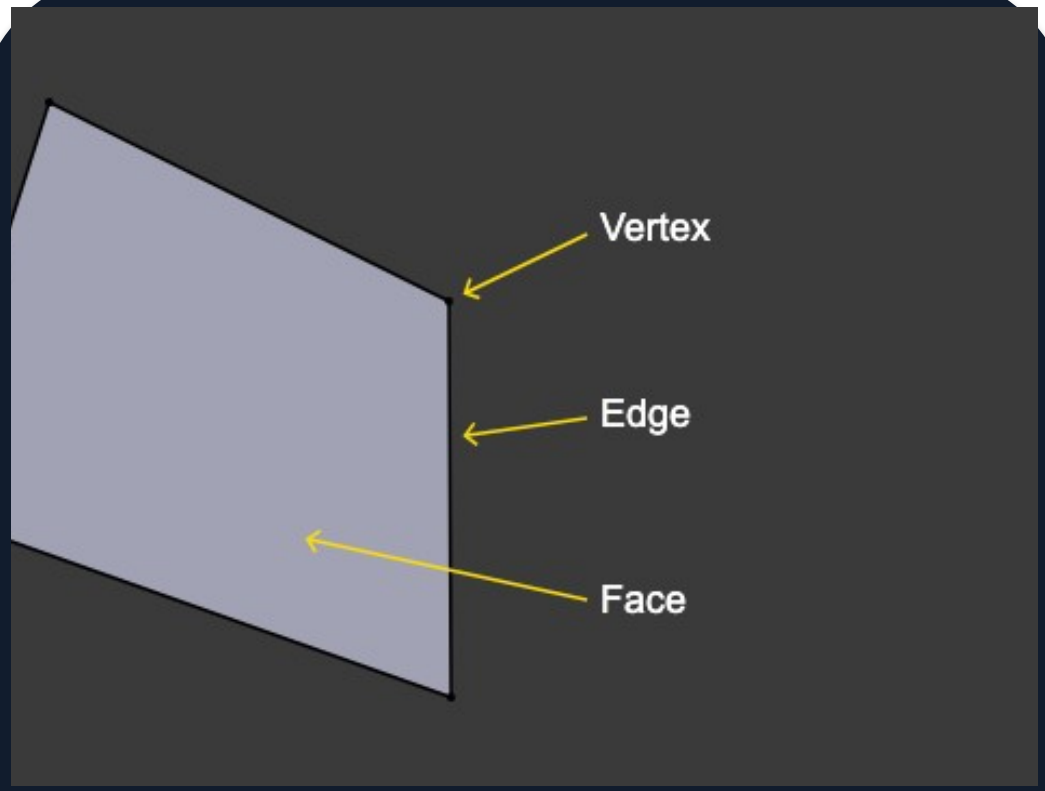
- Add loops only for form or deformation
- Place poles away from bending areas
- Inspect face orientation
- Prefer clean local repairs

FAILURE MODES TO CONTROL

- Non-manifold edges
- Flipped normals
- Dense, irregular topology

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Model with mesh anatomy and edge flow



LEGACY BLENDER REFERENCE · CURRENT WORKFLOW EXPLAINED AT RIGHT

PRINCIPLE

Vertices, edges and faces form loops that support silhouette, shading and deformation.

APPLIED EXAMPLE

Concentric eye loops support expression while longitudinal body loops preserve a smooth fish silhouette.

EVIDENCE TO TRACK

1. Manifold target met
2. Normals outward
3. Loops support animation

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Use Edit Mode tools as a controlled vocabulary

1

Extrude

Extend connected form

2

Inset

Create an inner boundary

3

Bevel

Control edge width and highlights

4

Loop Cut

Add resolution around a continuous loop

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Decisions and risks — Use Edit Mode tools as a controlled vocabulary

DECISIONS THAT IMPROVE QUALITY

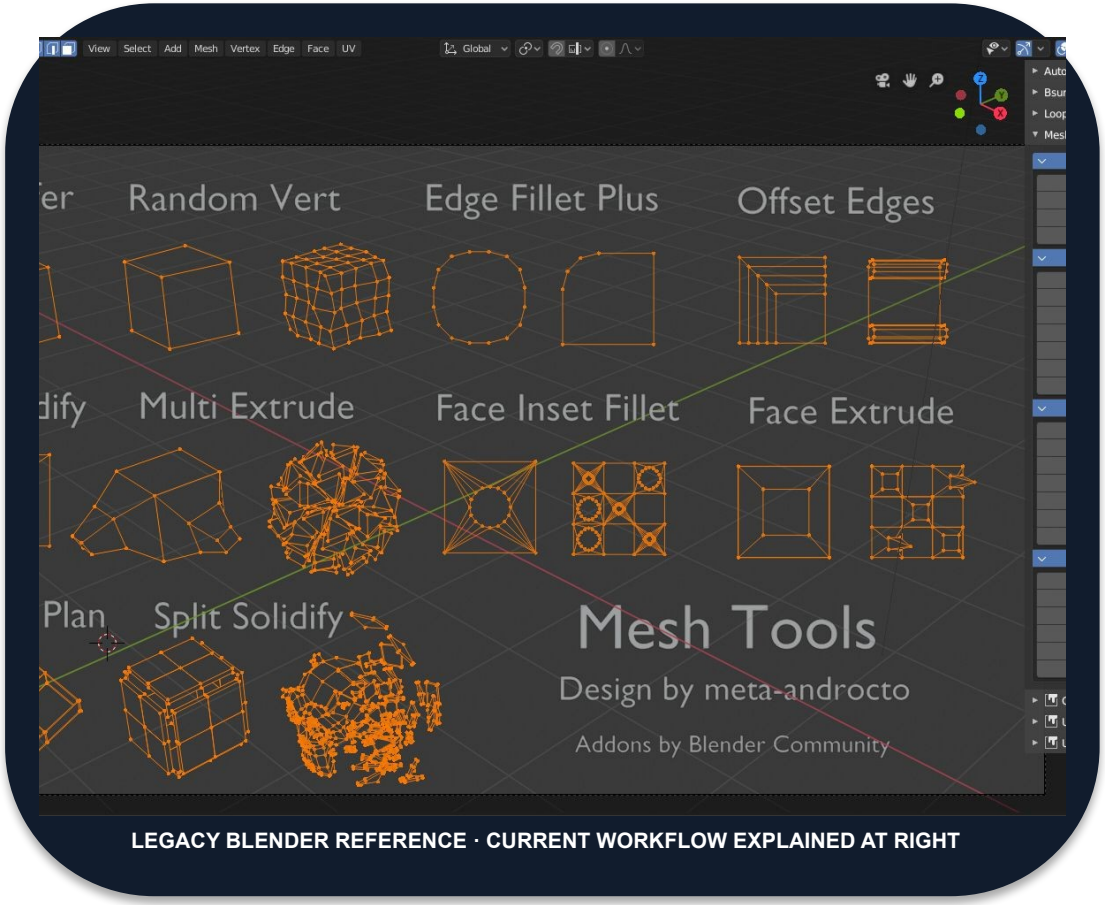
- Extrude along a constrained direction
- Avoid stacked accidental extrusions
- Use bevel width consistently
- Add cuts where curvature needs support

FAILURE MODES TO CONTROL

- Duplicate faces from cancelled moves
- Bevel pinching
- Loop cut stops at poles

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Use Edit Mode tools as a controlled vocabulary



LEGACY BLENDER REFERENCE · CURRENT WORKFLOW EXPLAINED AT RIGHT

PRINCIPLE

Extrude, inset, bevel and loop cut should each answer a form or topology need.

APPLIED EXAMPLE

The tail grows from an extrusion, while a small bevel produces a readable highlight without excessive subdivision.

EVIDENCE TO TRACK

1. No duplicate geometry
2. Curvature reads cleanly
3. Budget remains controlled

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CONCEPT VISUAL

Mesh quality — before and after



The same conceptual object can look similar in silhouette while differing radically in topology. Validate manifold status, edge flow and shading before approval.

Concept visual — validate against the editable Blender file.

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Sculpt from broad forms to controlled detail

1

Brushes

Grab, clay, smooth and crease for distinct jobs

2

Symmetry

Preserve paired form while blocking

3

Dyntopo

Local dynamic detail with topology cost

4

Remesh

Redistribute resolution for sculpting

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Decisions and risks — Sculpt from broad forms to controlled detail

DECISIONS THAT IMPROVE QUALITY

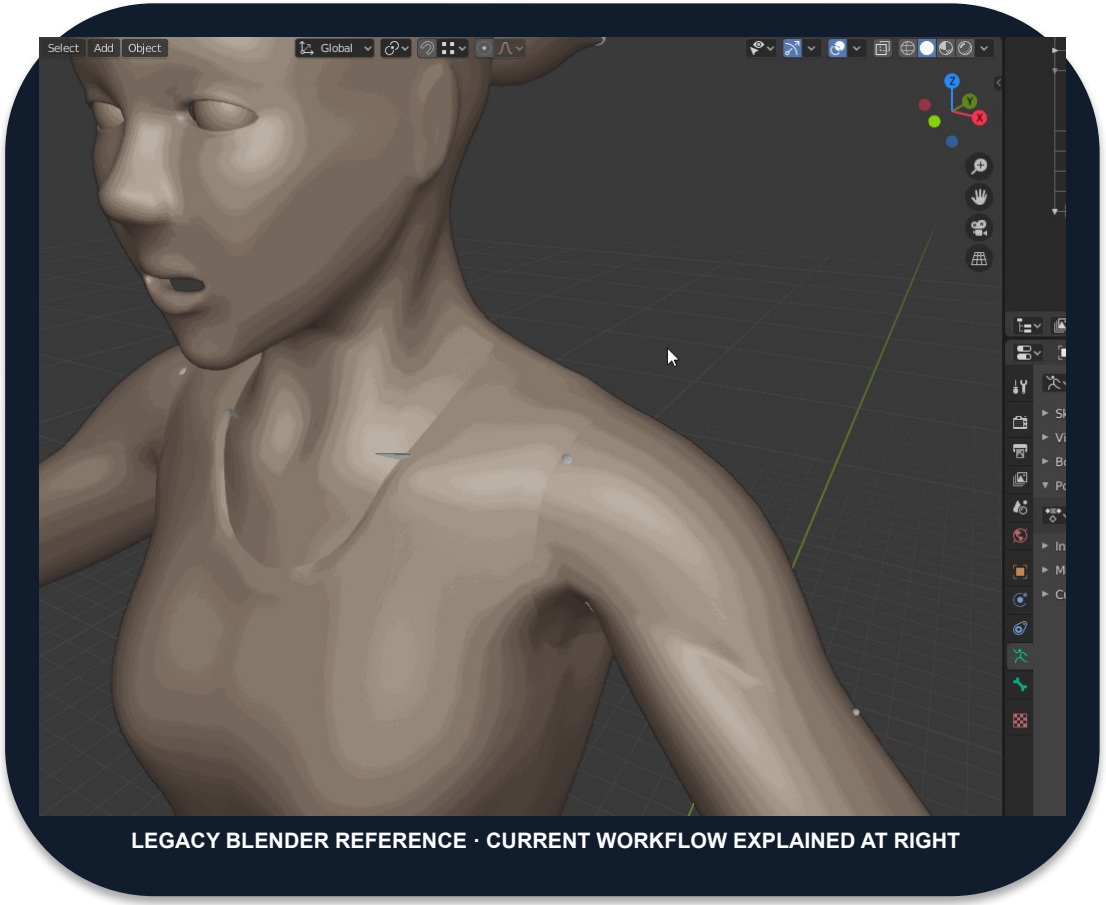
- Lock primary forms before detail
- Choose resolution for the current stage
- Save before remesh
- Retopologise if animation demands it

FAILURE MODES TO CONTROL

- Detail added at the wrong scale
- Dyntopo destroys planned topology
- Over-smoothing removes landmarks

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Sculpt from broad forms to controlled detail



LEGACY BLENDER REFERENCE · CURRENT WORKFLOW EXPLAINED AT RIGHT

PRINCIPLE

Sculpting is most predictable when resolution, symmetry and brush purpose are managed.

APPLIED EXAMPLE

Grab establishes the fish belly and tail arc; Smooth removes noise; Crease defines the mouth only after proportions pass.

EVIDENCE TO TRACK

1. Primary form approved
2. Symmetry intentional
3. Topology plan stated

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Review and refine the prototype

1

Compare

Overlay reference and inspect key views

2

Measure

Dimensions, transforms and polygon statistics

3

Inspect

Normals, manifold status and modifier result

4

Decide

Accept, repair or escalate with reasons

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Decisions and risks — Review and refine the prototype

DECISIONS THAT IMPROVE QUALITY

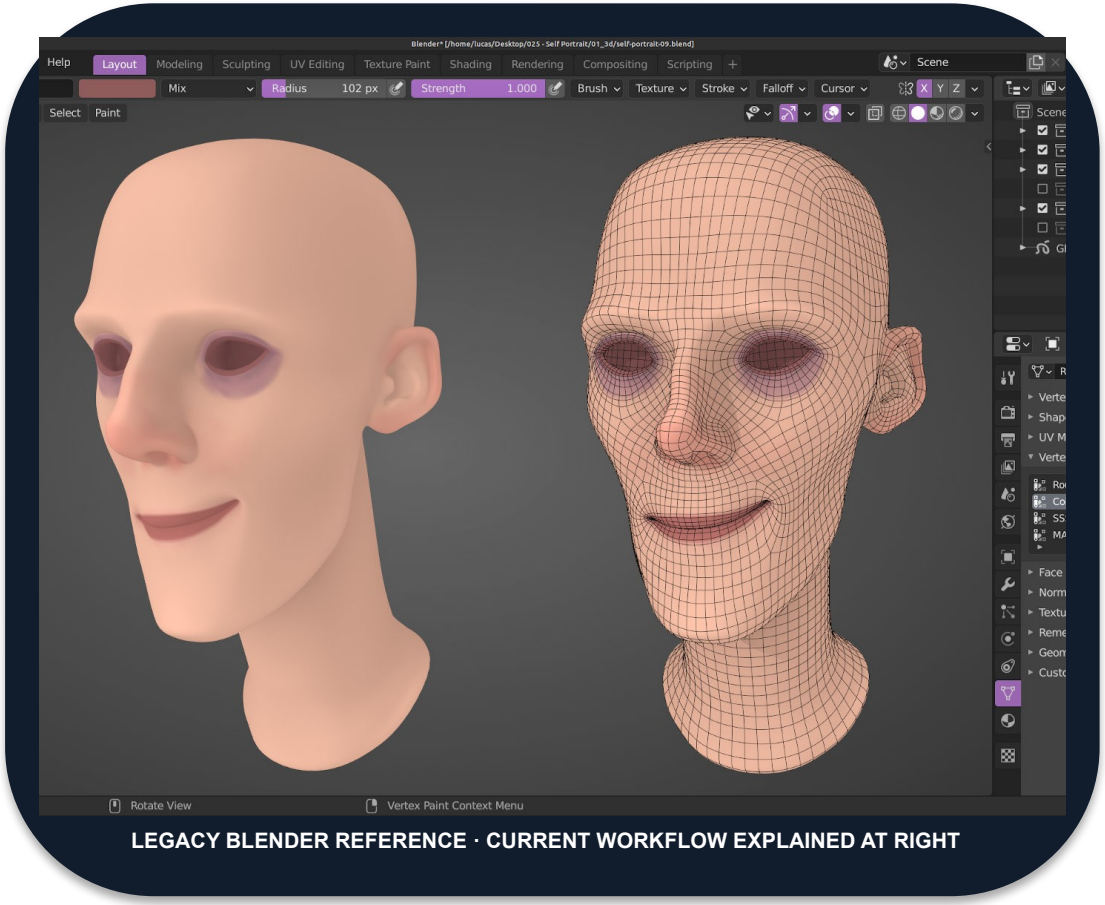
- Preserve a baseline
- Review one failure class at a time
- Capture before/after evidence
- Record remaining limitations

FAILURE MODES TO CONTROL

- Changing many variables at once
- Review only the beauty render
- No trace of rejected versions

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Review and refine the prototype



LEGACY BLENDER REFERENCE · CURRENT WORKFLOW EXPLAINED AT RIGHT

PRINCIPLE

A prototype is accepted through criteria, evidence and iteration—not appearance alone.

APPLIED EXAMPLE

The goldfish passes silhouette and 500-polygon criteria after a local tail repair and reduction of hidden underside loops.

EVIDENCE TO TRACK

1. Acceptance checklist complete
2. Defects closed or owned
3. Version and reviewer recorded

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HANDS-ON PRACTICE

Topic 2 labs

Lab 3

- Model and sculpt the production goldfish
- A recognisable goldfish model with intentional form, topology and saved milestones.

Lab 4

- Repair topology, normals and deformation zones
- A repaired, review-ready mesh with a defect log and before/after evidence.

Progression

- Inspect and plan
- Build and refine
- Verify and present

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Model and sculpt the production goldfish

LAB 3

The approved blockout needs clearer anatomy and controlled detail without losing its low-cost structure.

DELIVERABLE

A recognisable goldfish model with intentional form, topology and saved milestones.

ALIGNMENT

K3–K4 · A3

OBJECTIVE

Develop primary and secondary forms with suitable Edit Mode and sculpting tools.

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Lab 3 — example evidence



Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.

Concept visual — validate against the editable Blender file.

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ACCEPTANCE CRITERIA

Lab 3 — quality gates

Form

- Primary and secondary forms are readable
- Symmetry is intentional

Structure

- Detail level matches the stage
- Topology plan supports later animation

Delivery

- Milestone files and metrics are preserved
- A recognisable goldfish model with intentional form, topology and saved milestones.

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VERIFY, DOCUMENT, SUBMIT

Lab 3 — acceptance evidence

ACCEPTANCE CHECKS

- Primary and secondary forms are readable
- Symmetry is intentional
- Detail level matches the stage
- Topology plan supports later animation
- Milestone files and metrics are preserved

EVIDENCE TO SUBMIT

- Landmark view
- Sculpting before/after
- Wireframe view
- Polygon statistics
- Decision explanation

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Repair topology, normals and deformation zones

LAB 4

A review finds shading defects, dense areas and weak edge flow around the tail.

DELIVERABLE

A repaired, review-ready mesh with a defect log and before/after evidence.

ALIGNMENT

K4, K7 · A4

OBJECTIVE

Diagnose and repair the supplied mesh, then prove the revised prototype meets its criteria.

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Lab 4 — example evidence



Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.

Concept visual — validate against the editable Blender file.

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ACCEPTANCE CRITERIA

Lab 4 — quality gates

Form

- No unintended non-manifold geometry remains
- Normals face outward

Structure

- Silhouette remains approved
- Tail deformation holds volume

Delivery

- Defect log links each issue to evidence and result
- A repaired, review-ready mesh with a defect log and before/after evidence.

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VERIFY, DOCUMENT, SUBMIT

Lab 4 — acceptance evidence

ACCEPTANCE CHECKS

- No unintended non-manifold geometry remains
- Normals face outward
- Silhouette remains approved
- Tail deformation holds volume
- Defect log links each issue to evidence and result

EVIDENCE TO SUBMIT

- Baseline wireframe
- Face-orientation view
- Defect log
- Deformation test
- Final statistics

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WHAT YOU CAN NOW DEMONSTRATE

Topic 2 recap

1

Concepts

5 decision-ready frameworks

2

Labs

Labs 3–4 completed

3

Evidence

Editable Blender files, metrics and presentations

4

Alignment

LO2 · K3–K4, K7 · A3–A4

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TOPIC 3

03

Animate and Present the 3D Prototype

Rig movement · keyframe action · surface and present

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PROGRESSIVE LEARNING

Topic 3 — capability map

1

Understand

Rig movement

2

Apply

keyframe action

3

Review

LO3 · K5–K8 · A5–A6

4

Demonstrate

Labs 5–6

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Apply animation principles to readable motion

1

Timing

Frames allocated to an action

2

Spacing

Distance travelled between frames

3

Arcs

Natural curved motion paths

4

Overlap

Fins and tail follow the body

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Decisions and risks — Apply animation principles to readable motion

DECISIONS THAT IMPROVE QUALITY

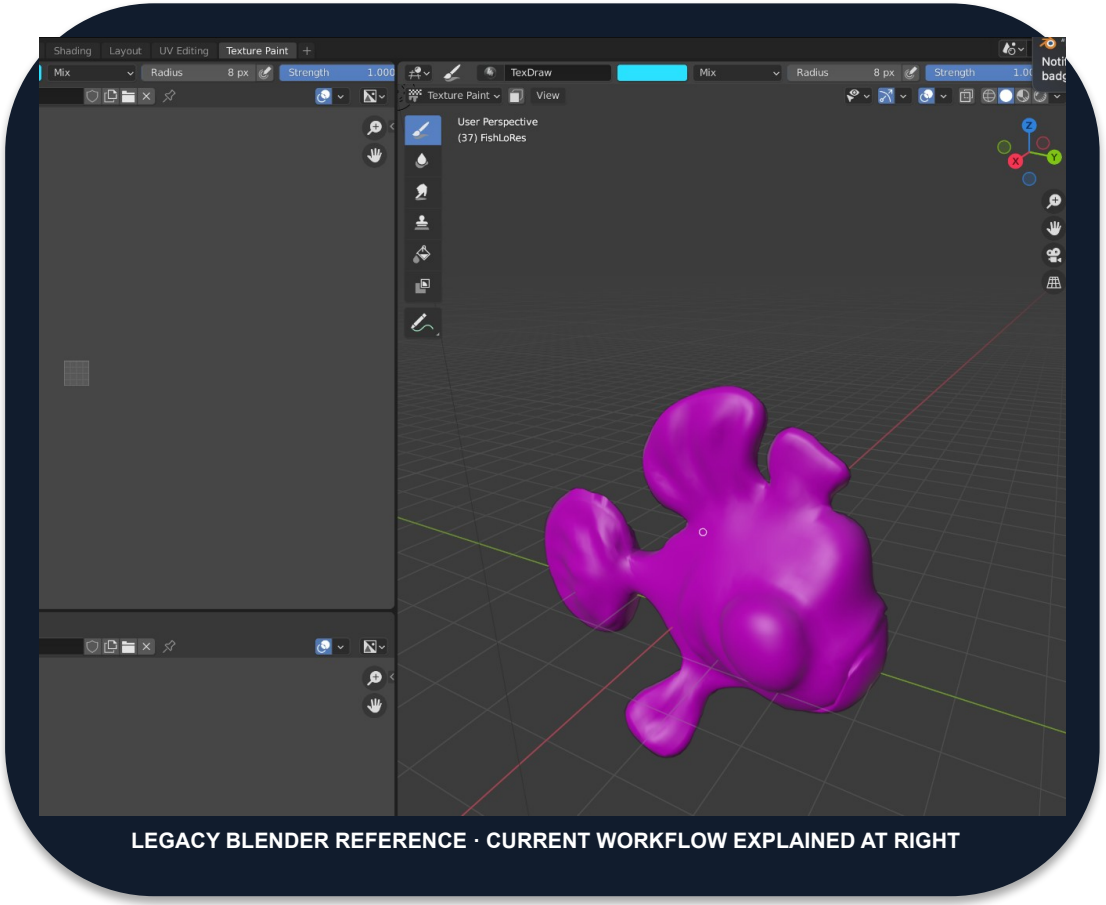
- Block key poses first
- Choose frame rate before animation
- Exaggerate only for readability
- Review motion at real speed

FAILURE MODES TO CONTROL

- Even spacing feels robotic
- No anticipation
- Secondary motion competes with the action

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Apply animation principles to readable motion



LEGACY BLENDER REFERENCE · CURRENT WORKFLOW EXPLAINED AT RIGHT

PRINCIPLE

Timing, spacing, anticipation and follow-through make mechanical keyframes feel intentional.

APPLIED EXAMPLE

A swim cycle leads with the body, then offsets the tail and fins to create overlap and a continuous arc.

EVIDENCE TO TRACK

1. Cycle loops cleanly
2. Pose silhouette readable
3. Timing matches brief

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Rig with armatures and controlled weights

1

Bones

Joint chain and local axes

2

Hierarchy

Parent-child motion relationships

3

Skinning

Bind mesh to the armature

4

Weights

Control vertex influence

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Decisions and risks — Rig with armatures and controlled weights

DECISIONS THAT IMPROVE QUALITY

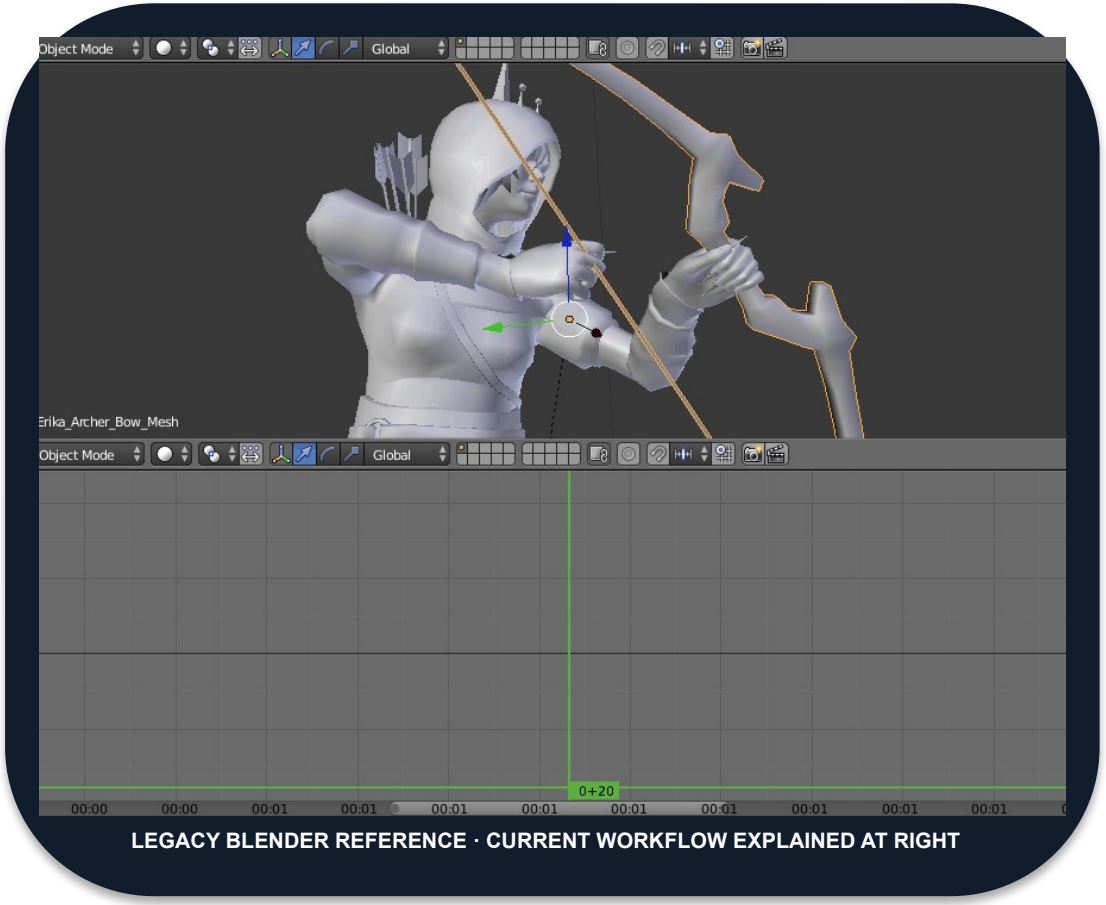
- Place joints at meaningful bend zones
- Check bone roll
- Test extreme poses
- Repair weights locally

FAILURE MODES TO CONTROL

- Bone axes inconsistent
- Weights leak across the body
- Topology collapses at a bend

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Rig with armatures and controlled weights



PRINCIPLE

A rig is a deformation system whose hierarchy and weights must match intended movement.

APPLIED EXAMPLE

A simple body-to-tail bone chain bends the fish while fin bones provide subtle independent motion.

EVIDENCE TO TRACK

1. Extreme poses hold volume
2. No unintended vertices move
3. Hierarchy is named

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Control keyframes, interpolation and curves

1

Keyframes

Location, rotation, scale or property states

2

Interpolation

Constant, linear or Bézier transitions

3

F-Curves

Editable value-over-time graphs

4

Dope Sheet

Timing and keyframe organisation

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Decisions and risks — Control keyframes, interpolation and curves

DECISIONS THAT IMPROVE QUALITY

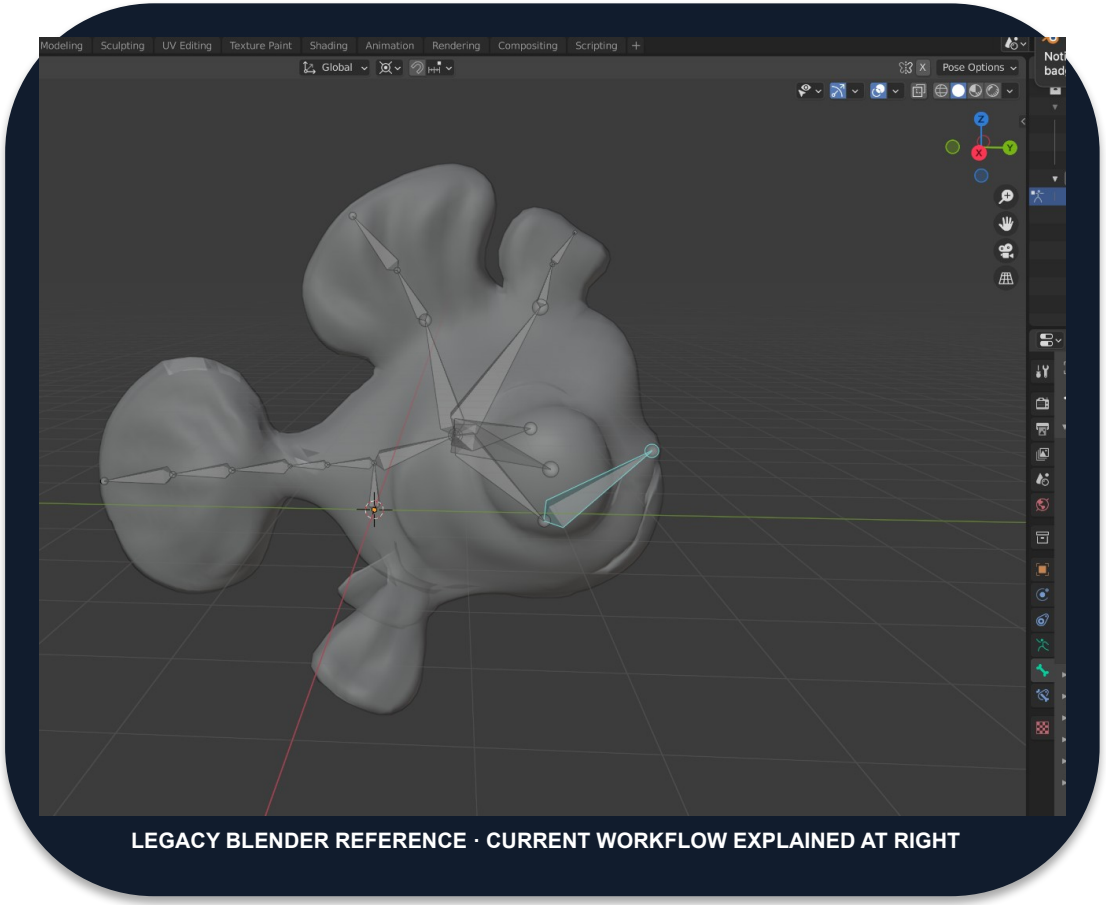
- Key only intended channels
- Use linear for constant motion
- Edit handles to remove overshoot
- Cycle only seamless channels

FAILURE MODES TO CONTROL

- Unwanted channels keyed
- Bézier overshoot
- Loop boundary jumps

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Control keyframes, interpolation and curves



LEGACY BLENDER REFERENCE · CURRENT WORKFLOW EXPLAINED AT RIGHT

PRINCIPLE

Keyframes store states; interpolation and F-curves define how the values change.

APPLIED EXAMPLE

The fish translates linearly across the bowl while tail rotation uses smooth cyclic curves with offset timing.

EVIDENCE TO TRACK

1. No accidental keys
2. Curve shape justified
3. Loop boundary matches

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Build materials from shaders, UVs and textures

1

Shader

Principled BSDF response to light

2

Channels

Base colour, roughness, metallic and normal

3

UV

2D coordinates with seams and islands

4

Texture

Image data with colour-space choices

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Decisions and risks — Build materials from shaders, UVs and textures

DECISIONS THAT IMPROVE QUALITY

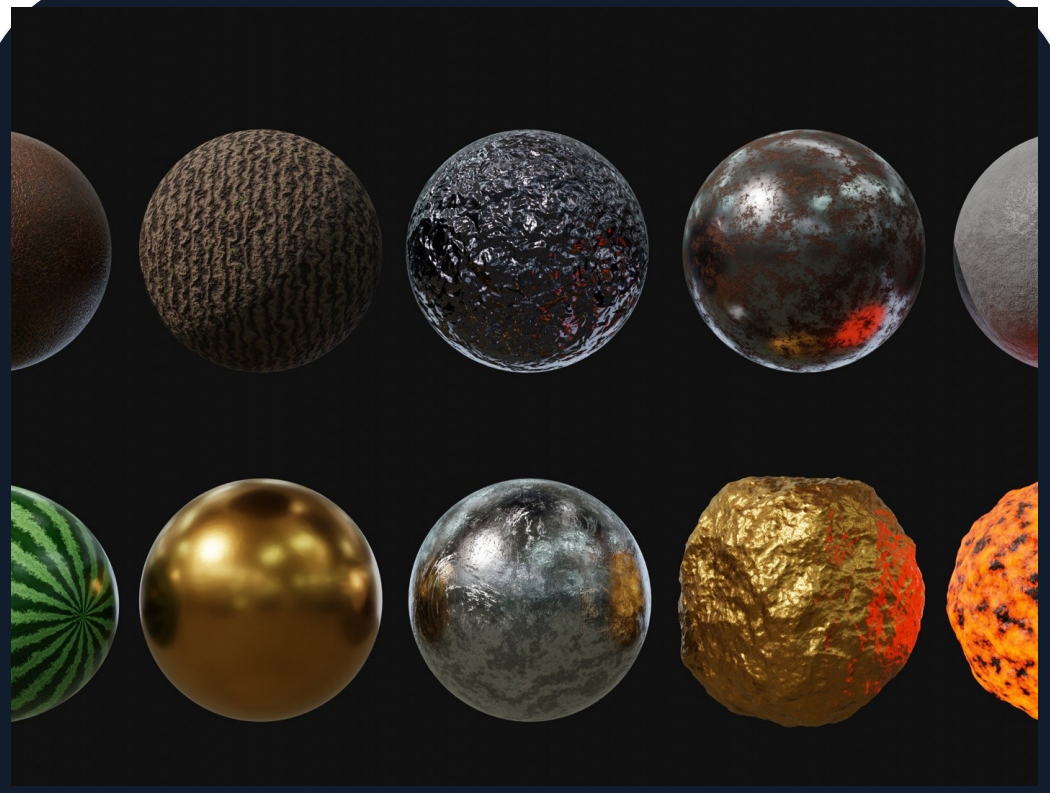
- Set scale before unwrapping
- Place seams in low-visibility areas
- Use Non-Color for data maps
- Test under neutral light

FAILURE MODES TO CONTROL

- Overlapping UVs unintentionally
- Wrong colour space
- Material judged under dramatic light only

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Build materials from shaders, UVs and textures



LEGACY BLENDER REFERENCE · CURRENT WORKFLOW EXPLAINED AT RIGHT

PRINCIPLE

Materials describe surface response; textures provide mapped data; UVs define how 2D data reaches the mesh.

APPLIED EXAMPLE

The fish uses a clean body UV, colour texture, moderate roughness and a subtle normal map that holds under rotation.

EVIDENCE TO TRACK

1. No unintended overlaps
2. Texel density consistent
3. Channels connected correctly

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Present the prototype for useful feedback

1

Context

Brief and target use

2

Views

Neutral orthographic and perspective evidence

3

Motion

Short loop at intended speed

4

Decision

Pass, limitation and requested approval

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Decisions and risks — Present the prototype for useful feedback

DECISIONS THAT IMPROVE QUALITY

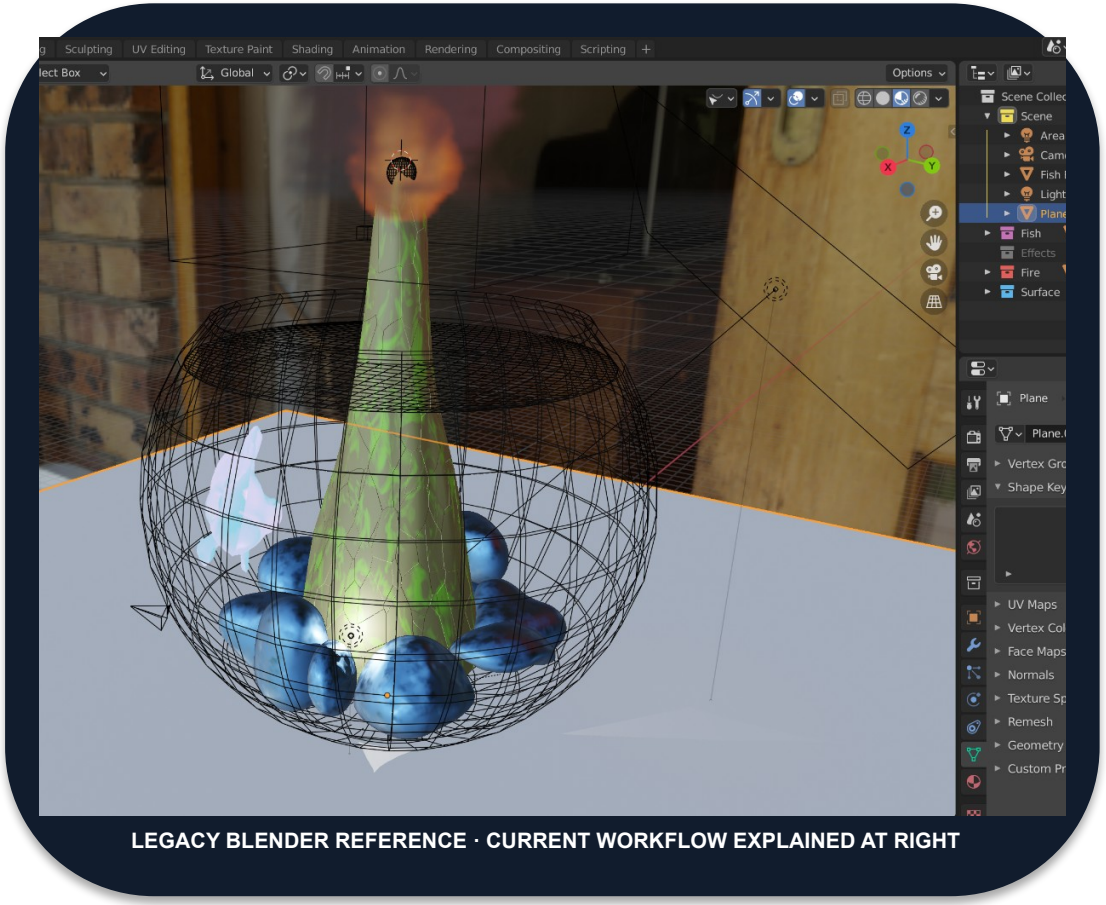
- Use comparable before/after views
- Show statistics and version
- State open risks
- Ask a specific review question

FAILURE MODES TO CONTROL

- Only a beauty render
- No version identification
- Feedback request too broad

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Present the prototype for useful feedback



LEGACY BLENDER REFERENCE · CURRENT WORKFLOW EXPLAINED AT RIGHT

PRINCIPLE

A review presentation should expose intent, process, evidence and the decision required.

APPLIED EXAMPLE

The review pack shows the concept, wireframe, UV/material view and swim cycle, then requests approval to proceed to final lighting.

EVIDENCE TO TRACK

1. All required views present
2. Decision owner named
3. Feedback logged

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HANDS-ON PRACTICE

Topic 3 labs

Lab 5

- UV unwrap and surface the goldfish
- A surfaced fish with validated UV layout, material channels and texture dependencies.

Lab 6

- Rig and animate a looping swim cycle
- A rigged fish with a looped swim animation and deformation evidence.

Progression

- Inspect and plan
- Build and refine
- Verify and present

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UV unwrap and surface the goldfish

LAB 5

The approved mesh needs a stable PBR material for close review and animation.

DELIVERABLE

A surfaced fish with validated UV layout, material channels and texture dependencies.

ALIGNMENT

K6–K7 · A5

OBJECTIVE

Create controlled UVs, textures and a Principled BSDF material that remains readable under neutral light.

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Lab 5 — example evidence



Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.

Concept visual — validate against the editable Blender file.

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ACCEPTANCE CRITERIA

Lab 5 — quality gates

Form

- UVs have no unintended overlaps
- Checker distortion is controlled

Structure

- Material channels use correct colour spaces
- Surface reads under neutral lighting

Delivery

- Texture dependencies are included
- A surfaced fish with validated UV layout, material channels and texture dependencies.

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Lab 5 — acceptance evidence

ACCEPTANCE CHECKS

- UVs have no unintended overlaps
- Checker distortion is controlled
- Material channels use correct colour spaces
- Surface reads under neutral lighting
- Texture dependencies are included

EVIDENCE TO SUBMIT

- UV layout
- Checker test
- Shader node graph
- Neutral render
- Dependency inventory

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Rig and animate a looping swim cycle

LAB 6

The surfaced fish must perform a short, readable swim cycle for stakeholder review.

DELIVERABLE

A rigged fish with a looped swim animation and deformation evidence.

ALIGNMENT

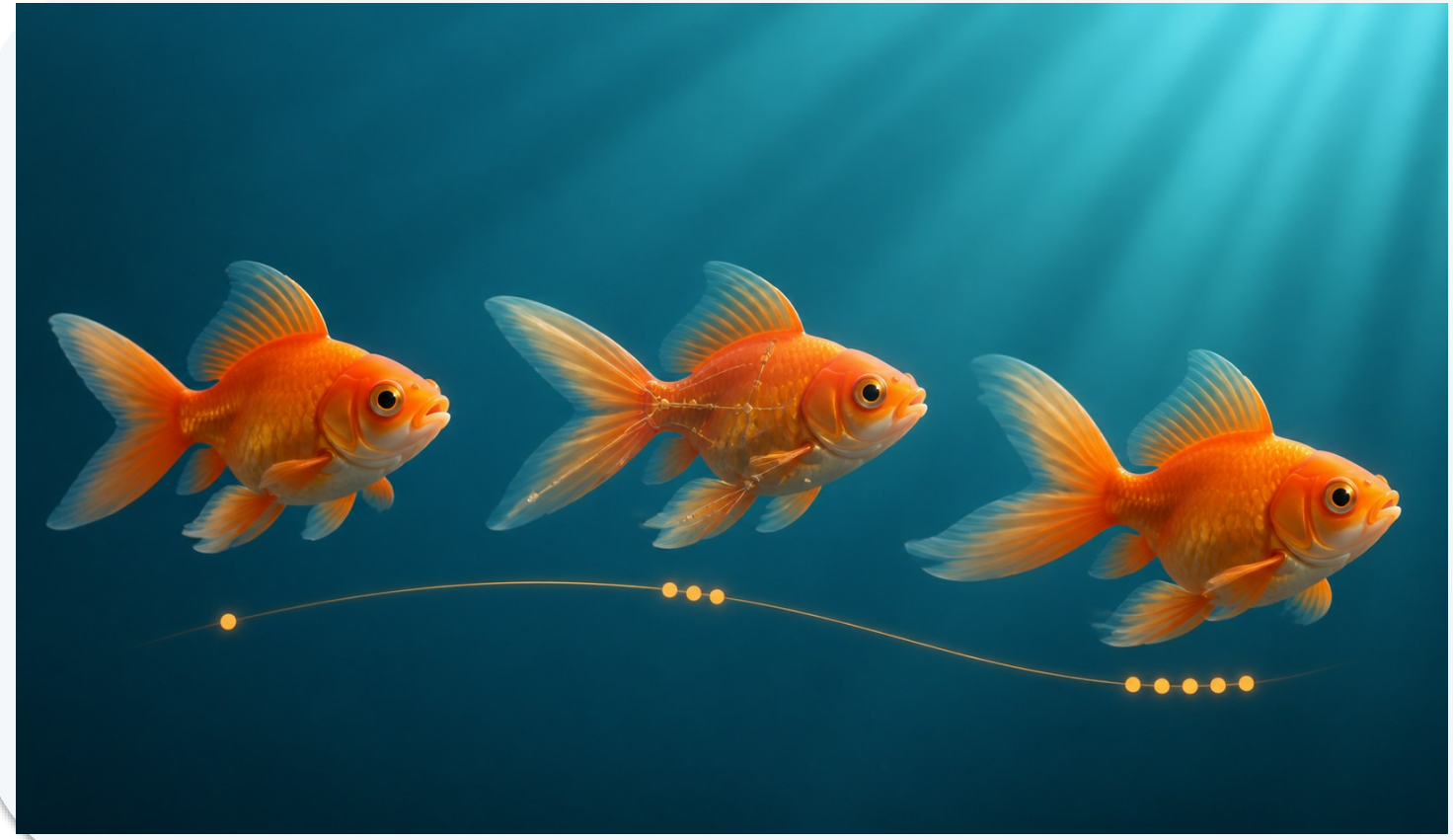
K5–K6 · A5–A6

OBJECTIVE

Build a simple armature, validate deformation and animate a clean loop using controlled curves.

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Lab 6 — example evidence



Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.

Concept visual — validate against the editable Blender file.

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ACCEPTANCE CRITERIA

Lab 6 — quality gates

Form

- Rig hierarchy and bones are named
- Extreme poses hold volume

Structure

- No unintended vertices move
- Animation loop is seamless

Delivery

- Timing and interpolation match the brief
- A rigged fish with a looped swim animation and deformation evidence.

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Lab 6 — acceptance evidence

ACCEPTANCE CHECKS

- Rig hierarchy and bones are named
- Extreme poses hold volume
- No unintended vertices move
- Animation loop is seamless
- Timing and interpolation match the brief

EVIDENCE TO SUBMIT

- Armature hierarchy
- Weight-paint view
- Extreme-pose test
- F-curve view
- Loop playback or frame comparison

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WHAT YOU CAN NOW DEMONSTRATE

Topic 3 recap

1

Concepts

5 decision-ready frameworks

2

Labs

Labs 5–6 completed

3

Evidence

Editable Blender files, metrics and presentations

4

Alignment

LO3 · K5–K8 · A5–A6

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TOPIC 4

04

Optimise, Render and Export the Model

Frame the asset · validate quality · deliver reliably

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PROGRESSIVE LEARNING

Topic 4 — capability map

1

Understand

Frame the asset

2

Apply

validate quality

3

Review

LO4 · K8–K9 · A7–A8

4

Demonstrate

Labs 7–8

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Use cameras to control what the audience sees

1

Projection

Perspective or orthographic

2

Lens

Field of view and spatial compression

3

Composition

Focal point, balance and negative space

4

DOF

Focus plane and controlled blur

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Decisions and risks — Use cameras to control what the audience sees

DECISIONS THAT IMPROVE QUALITY

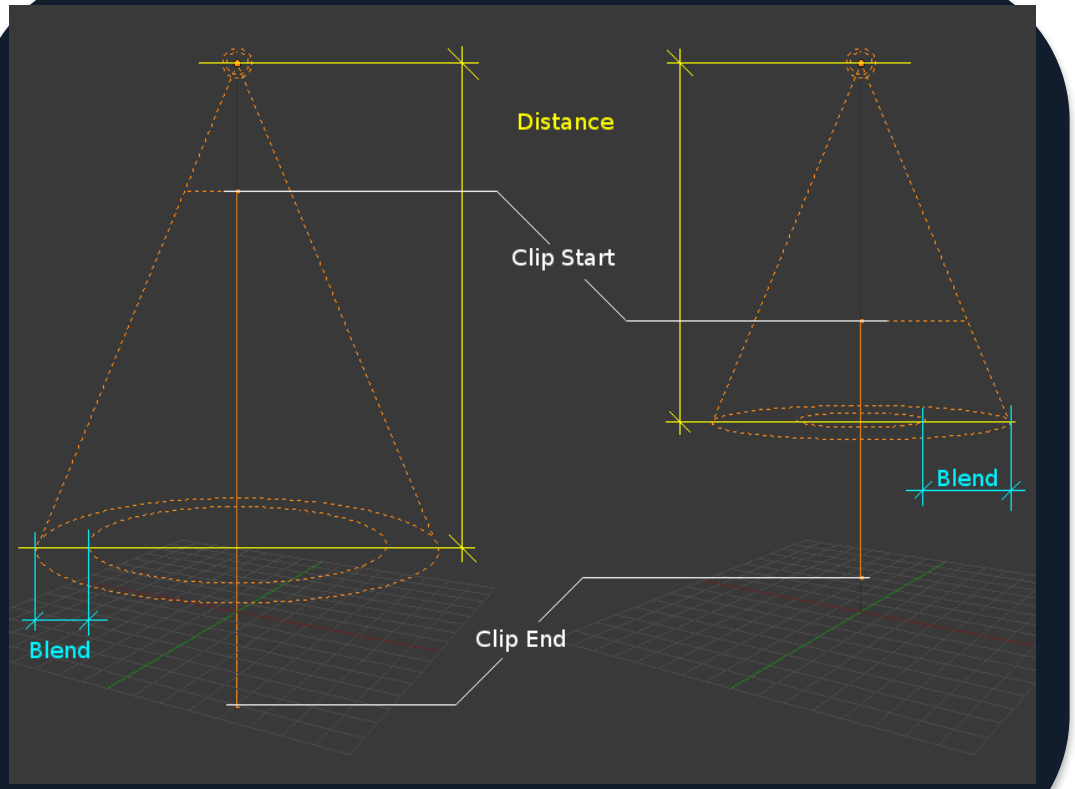
- Choose projection for the deliverable
- Avoid lens distortion that changes review judgement
- Lock approved cameras
- Use DOF sparingly in evidence views

FAILURE MODES TO CONTROL

- Wide lens distorts form
- Clipped geometry
- Beauty DOF hides defects

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Use cameras to control what the audience sees



LEGACY BLENDER REFERENCE · CURRENT WORKFLOW EXPLAINED AT RIGHT

PRINCIPLE

Camera projection, lens, composition and depth of field must serve the communication goal.

APPLIED EXAMPLE

A 50 mm perspective camera presents the fish naturally; orthographic cameras remain for proportion evidence.

EVIDENCE TO TRACK

1. Subject fully framed
2. Evidence views distortion-free
3. Camera named and locked

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Light form with purpose

1

Key

Primary form-defining light

2

Fill

Controls shadow contrast

3

Rim

Separates silhouette from background

4

World

Ambient environment contribution

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Decisions and risks — Light form with purpose

DECISIONS THAT IMPROVE QUALITY

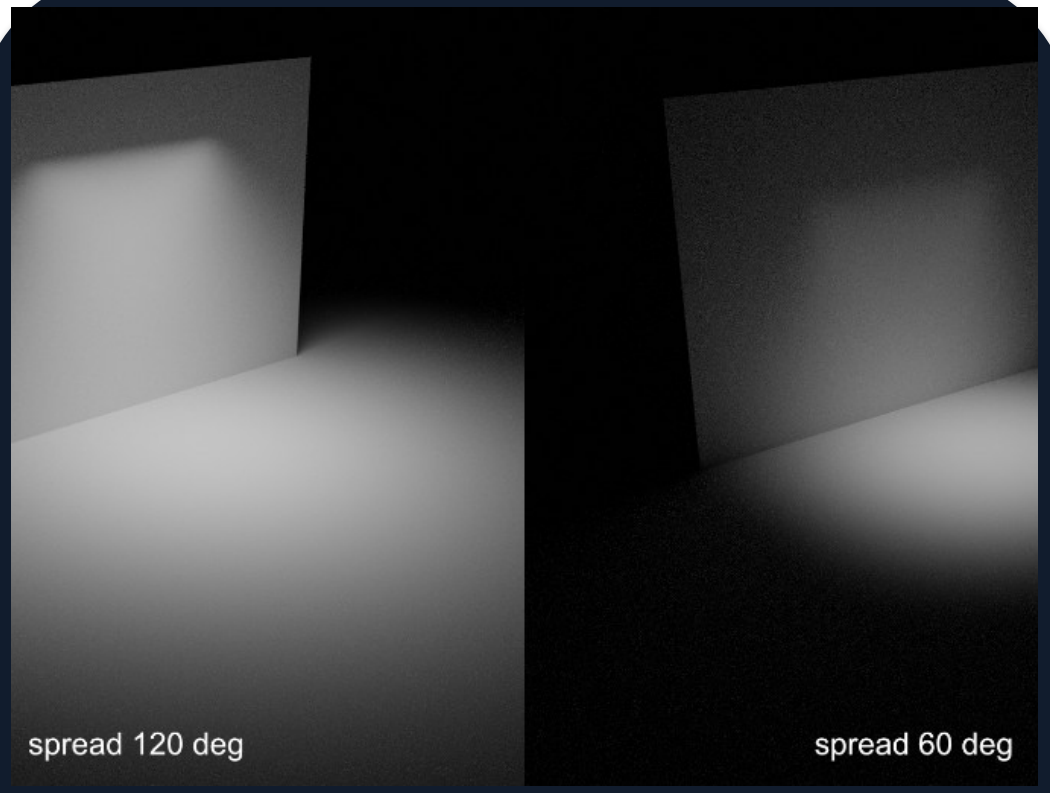
- Start with one light
- Use large sources for soft highlights
- Check colour management
- Separate presentation from diagnostic lighting

FAILURE MODES TO CONTROL

- Flat multi-light setup
- Blown highlights
- Lighting masks surface problems

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Light form with purpose



LEGACY BLENDER REFERENCE · CURRENT WORKFLOW EXPLAINED AT RIGHT

PRINCIPLE

Light direction, size, intensity and colour reveal shape while controlling contrast and render cost.

APPLIED EXAMPLE

A large area key reveals the body curve, a soft fill retains eye detail and a rim separates the tail.

EVIDENCE TO TRACK

1. Silhouette separated
2. Highlights retain detail
3. Neutral check render passes

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Choose a render engine and settings

1. **Workbench**

   Fast geometry and studio-light review

2. **Eevee**

   Real-time rasterised presentation

3. **Cycles**

   Path-traced lighting accuracy

4. **Output**

   Resolution, samples, format and colour

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Decisions and risks — Choose a render engine and settings

DECISIONS THAT IMPROVE QUALITY

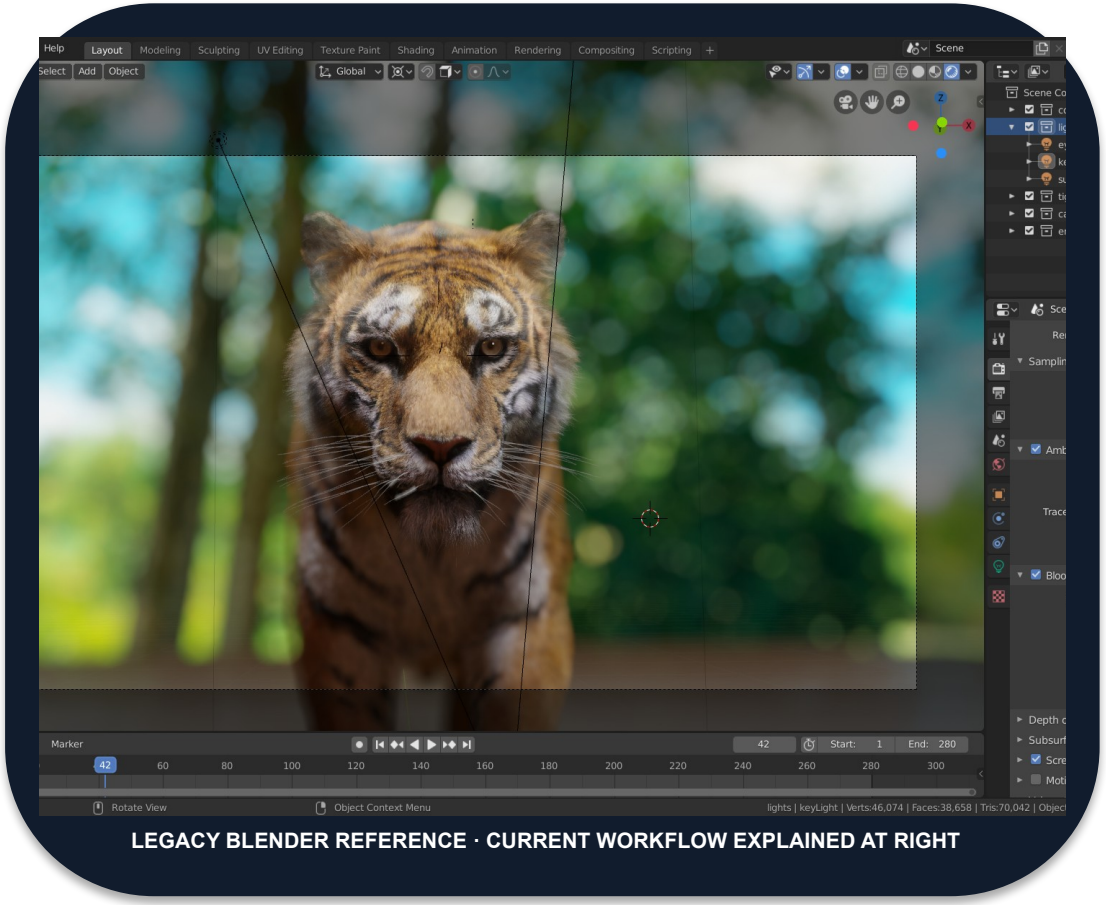
- Use Workbench for topology evidence
- Use Eevee for fast iteration
- Use Cycles when light transport matters
- Set output before final render

FAILURE MODES TO CONTROL

- Engine chosen by habit
- Unbounded samples
- Wrong colour or alpha output

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Choose a render engine and settings



PRINCIPLE

Workbench, Eevee and Cycles trade speed, realism and diagnostic value.

APPLIED EXAMPLE

The review uses Workbench wireframes, Eevee previews and a limited-sample Cycles final render.

EVIDENCE TO TRACK

1. Render purpose stated
2. Time within budget
3. Output settings verified

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Optimise the complete asset

1

Geometry

Delete hidden or duplicate elements

2

Transforms

Apply scale and confirm origin

3

Data

Remove unused materials and orphan data

4

Performance

Measure triangles, textures and render time

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Decisions and risks — Optimise the complete asset

DECISIONS THAT IMPROVE QUALITY

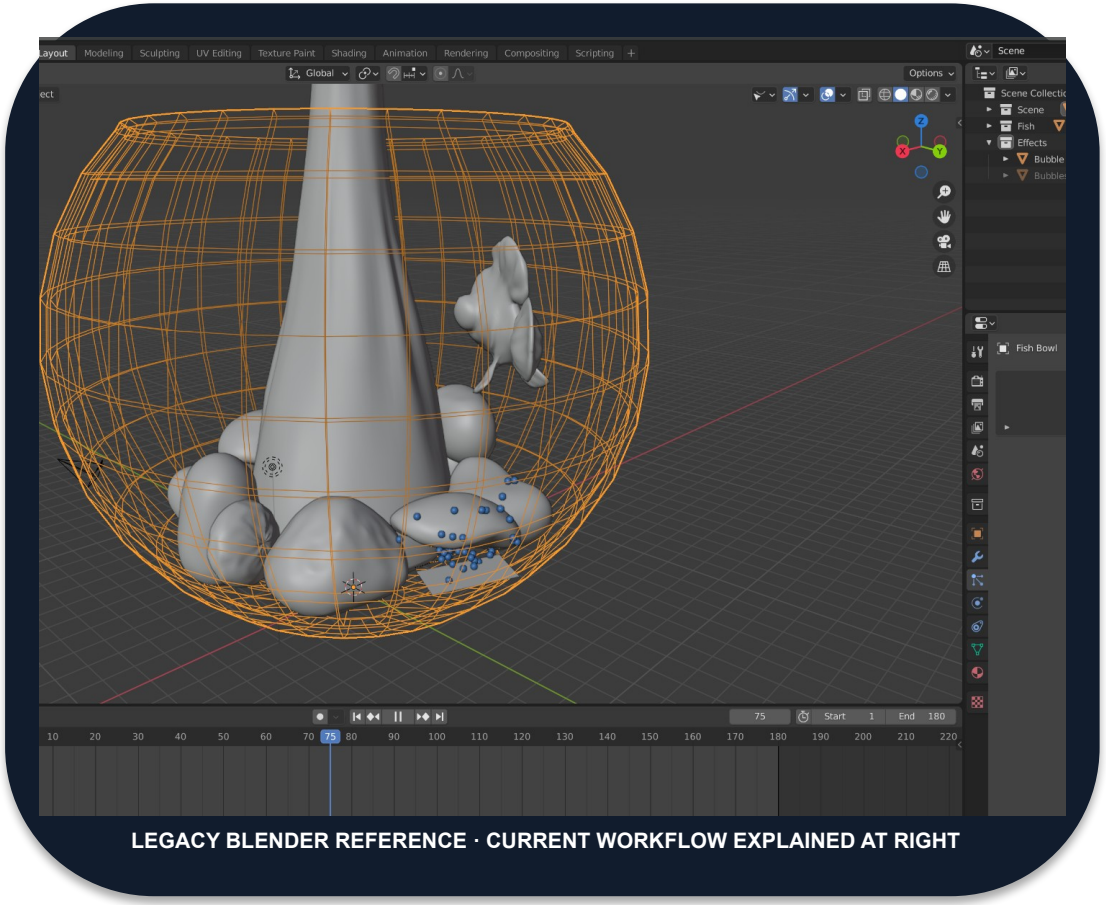
- Optimise against a measured baseline
- Protect deformation loops
- Test after each material change
- Keep the editable master

FAILURE MODES TO CONTROL

- Destructive decimation
- Missing texture dependency
- Optimisation changes animation

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Optimise the complete asset



LEGACY BLENDER REFERENCE · CURRENT WORKFLOW EXPLAINED AT RIGHT

PRINCIPLE

Optimisation removes cost without breaking silhouette, deformation, shading or required data.

APPLIED EXAMPLE

The final fish meets its target after hidden interior faces and redundant tail loops are removed, then the animation is retested.

EVIDENCE TO TRACK

1. Target budget met
2. Visual delta acceptable
3. Animation regression-free

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Export and verify the delivery package

1

GLB/glTF

Web and real-time PBR delivery

2

FBX

Animation and pipeline interchange

3

OBJ

Simple static mesh interchange

4

STL

Geometry-only fabrication workflow

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Decisions and risks — Export and verify the delivery package

DECISIONS THAT IMPROVE QUALITY

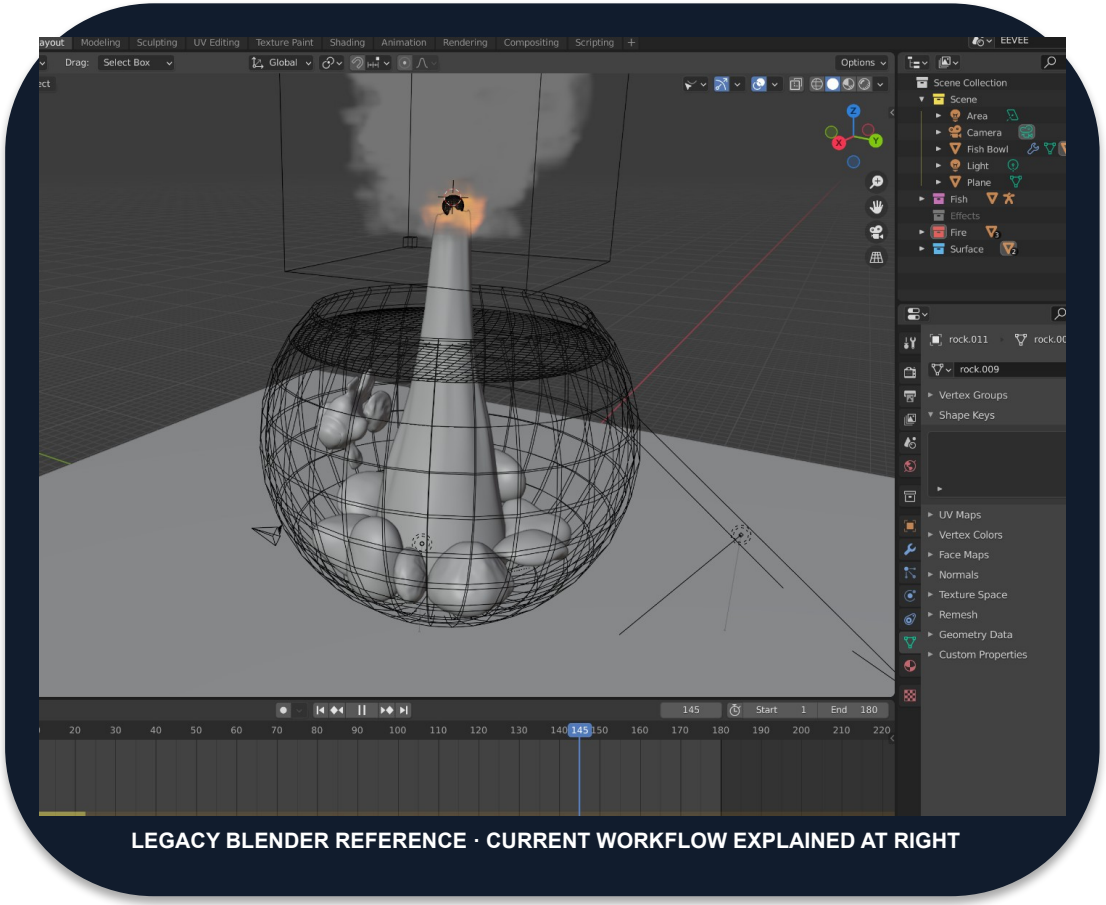
- Match axes, units and scale
- Export only intended objects
- Embed or package dependencies
- Re-import and compare

FAILURE MODES TO CONTROL

- Correct-looking source but broken export
- Missing textures
- Wrong axes or scale

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Export and verify the delivery package



PRINCIPLE

An export is complete only after round-trip validation in a clean environment.

APPLIED EXAMPLE

The approved fish is exported as GLB, reopened in a clean file and checked for dimensions, materials, hierarchy and animation.

EVIDENCE TO TRACK

1. Clean re-import passes
2. Dependencies present
3. Release manifest complete

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HANDS-ON PRACTICE

Topic 4 labs

Lab 7

- Create bubbles, light the scene and present for review
- A review scene with bubbles, three-point lighting and a decision-ready presentation.

Lab 8

- Optimise, render and export the fish-bowl capstone
- A release package containing editable source, GLB/FBX export, preview and validation manifest.

Progression

- Inspect and plan
- Build and refine
- Verify and present

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Create bubbles, light the scene and present for review

LAB 7

Stakeholders need a short aquarium presentation that keeps the fish readable and demonstrates controlled secondary motion.

DELIVERABLE

A review scene with bubbles, three-point lighting and a decision-ready presentation.

ALIGNMENT

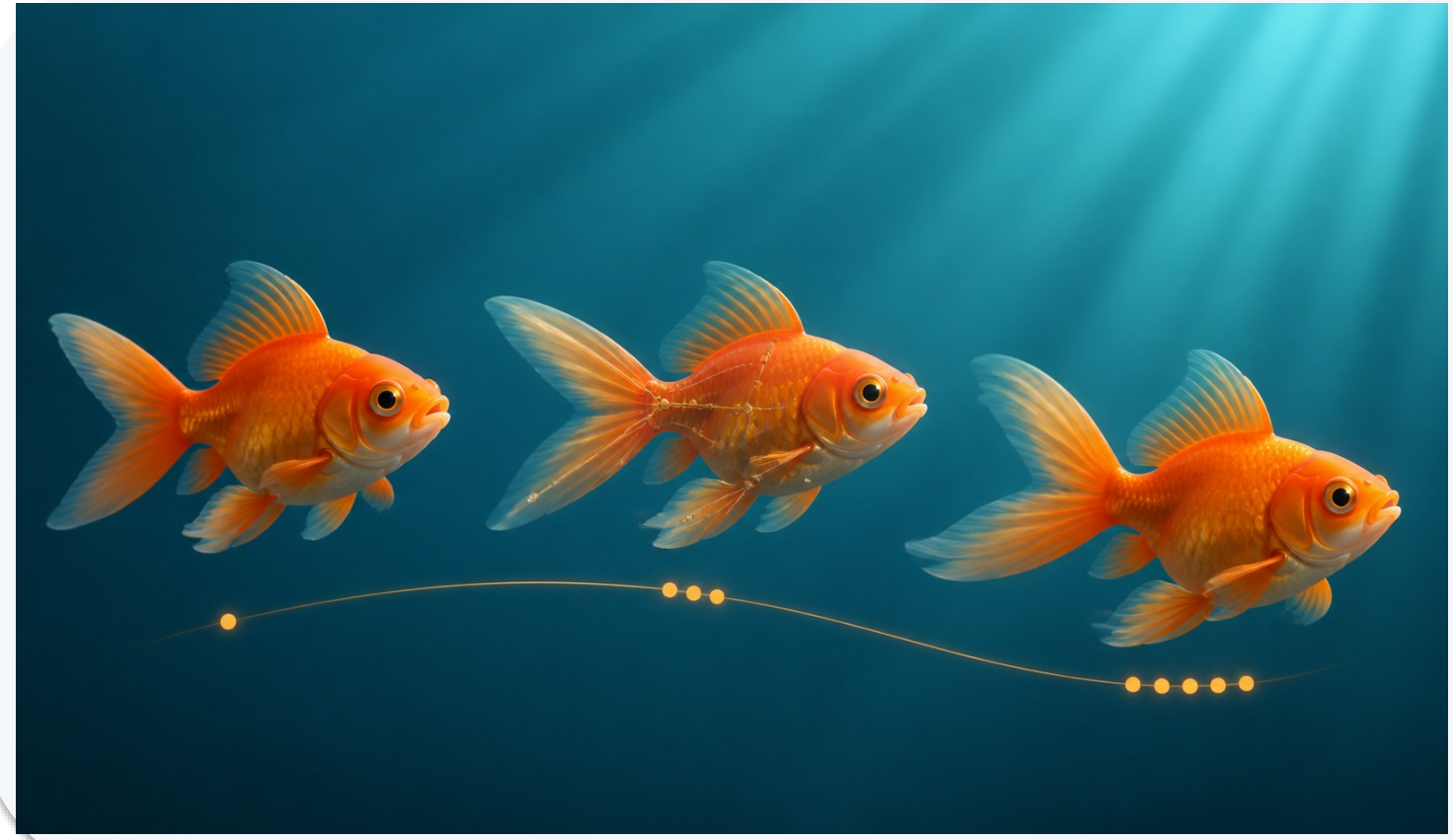
K8 · A6–A7

OBJECTIVE

Add a lightweight bubble/force effect, build purposeful lighting and present a review decision.

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Lab 7 — example evidence



Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.

Concept visual — validate against the editable Blender file.

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ACCEPTANCE CRITERIA

Lab 7 — quality gates

Form

- Secondary motion supports rather than hides the fish
- Simulation range is reproducible

Structure

- Lighting reveals silhouette and material
- Review evidence includes version and metrics

Delivery

- Feedback and decision are recorded
- A review scene with bubbles, three-point lighting and a decision-ready presentation.

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VERIFY, DOCUMENT, SUBMIT

Lab 7 — acceptance evidence

ACCEPTANCE CHECKS

- Secondary motion supports rather than hides the fish
- Simulation range is reproducible
- Lighting reveals silhouette and material
- Review evidence includes version and metrics
- Feedback and decision are recorded

EVIDENCE TO SUBMIT

- Particle settings
- Simulation frame
- Lighting diagram or view
- Review slide
- Decision log

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Optimise, render and export the fish-bowl capstone

LAB 8

The approved animated fish must be packaged for a lightweight viewer and handed off with verification evidence.

DELIVERABLE

A release package containing editable source, GLB/FBX export, preview and validation manifest.

ALIGNMENT

K8–K9 · A7–A8

OBJECTIVE

Optimise the full scene, render final evidence, export the intended asset and validate the package by round trip.

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Lab 8 — example evidence



Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.

Concept visual — validate against the editable Blender file.

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ACCEPTANCE CRITERIA

Lab 8 — quality gates

Form

- Final source remains editable
- Geometry and performance targets are met

Structure

- Final renders use verified settings
- Clean round-trip import passes

Delivery

- Manifest and package contain all dependencies
- A release package containing editable source, GLB/FBX export, preview and validation manifest.

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Lab 8 — acceptance evidence

ACCEPTANCE CHECKS

- Final source remains editable
- Geometry and performance targets are met
- Final renders use verified settings
- Clean round-trip import passes
- Manifest and package contain all dependencies

EVIDENCE TO SUBMIT

- Before/after statistics
- Neutral and final renders
- Export settings
- Clean re-import checks
- Release manifest

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Topic 4 recap

1

Concepts

5 decision-ready frameworks

2

Labs

Labs 7–8 completed

3

Evidence

Editable Blender files, metrics and presentations

4

Alignment

LO4 · K8–K9 · A7–A8

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COURSE WRAP-UP

05

Review, assessment and release

From creative brief to verified Blender asset

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LEARNING OUTCOMES

What you achieved

1

Analyse

Translate the creative brief into a polygon budget and criteria.

2

Develop

Model, sculpt and refine a structured Blender prototype.

3

Animate

Rig, surface, keyframe and present a readable sequence.

4

Optimise

Light, render, export and verify the delivery package.

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END-TO-END CAPABILITY

Production workflow

1. Brief and acceptance criteria
2. Blender blockout
3. Model, surface and animate
4. Validate and target export
5. Release package and approval

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WHY IT MATTERS

The accountable workflow

A creative idea becomes production-ready through modelling decisions, evidence and repeatable checks.

Release only what you can explain, validate and reproduce.

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Download course materials

1. Open LMS-TMS
2. Select this course
3. Download learner slides and guide
4. Open the Activities folder
5. Complete WA then PP

Open course materials on LMS-TMS

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DAY 2 · 3:30 PM–6:00 PM

Assessment reminder

WA

- 9 SAQs · K1–K9
- 60 minutes
- Open book

PP

- 4 Blender tasks · A1–A8
- 90 minutes
- Open book

Complete

- Assessment attendance
- Upload to LMS
- Sign summary record

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FINAL REMINDER

Assessment flow

1. TRAQOM / SSG attendance
2. Written Assessment (SAQ)
3. Practical Performance (PP)
4. Upload answers to LMS
5. Sign Assessment Summary Record

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Digital Attendance (Mandatory)

1

Scan

Use the QR code shown by the trainer or administrator.

2

Submit

Confirm the assessment attendance is recorded.

3

Complete

Sit the WA and PP within the declared timing.

4

Retain

Follow trainer instructions for assessment records.

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WHY IT MATTERS

Thank you

You are ready to model, animate, optimise and deliver 3D assets with Blender.

Build deliberately. Validate rigorously.

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